An Aggregate B-to-C Lower Bound from Revealed-Sacrifice Observation

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Abstract

The Goal-Frontier Maximization framework's central measure, $\mathrm{vol_P}$, quantifies capability *possession*: the potential optionality available to a collective. The experiential value $\mathrm{vol_P}$ is meant to protect, however, requires capability *exercise*: the actual realization of that optionality in the world. Direct measurement of exercised optionality ($\mathrm{vol_R}$) requires observational access to what the collective actually does—a system that in its extreme is a panopticon. We reject this method in favor of a privacy-minimal observation channel.

This paper introduces *revealed-sacrifice observation* and proves its central measurement theorem. Each voluntary trade event—an agent surrendering a benchmarked capability X in exchange for an unbenchmarked bundle Y—carries a rational-choice inequality revealing that Y's realized contribution is at least as large as X's benchmark magnitude. Aggregating across events with pairwise poset-independent bundles yields the Aggregate B-to-C Lower Bound theorem: for the subset U of capability-space covered by admissible trade events in a window W,

$$\mathrm{vol_R}^{U,[W]} \;\geq\; \sum_n \Delta\mathrm{vol_P}(X_n),$$

under assumptions S0 (valuation bounded by exercise), S1 (free choice), S2 (benchmark grounding), S4(a) (valuation additivity), and the genuine-exchange conditions of Definition 1. The bound is non-decreasing in observed admissible events, composes with the commitment protocol of [Lasser, 2026a] to produce a privacy-minimal public aggregate on distinct-category batches, and satisfies formal privacy properties (no interior access, discretization, commitment-layer composability). Two observation modalities—money-sacrifice and time-sacrifice—triangulate the same underlying substrate via cooperative-market exchange, with existential non-fungibility residuals establishing witness-conditional strict improvement of the joint aggregate. We also show that $\mathrm{vol_R}$ is *not self-balancing*: removing an unexercised capability leaves $\mathrm{vol_R}$ unchanged while $\mathrm{vol_P}$ contracts, so $\mathrm{vol_R}$ is a diagnostic, not an optimization target. The need-sufficiency filter that keeps below-sufficiency sacrifices out of the observation channel—required for S1 to hold across events—is developed in the companion paper [Lasser, 2026d], along with the gap decomposition, wireheading-consistent concentration signal, and residual class characterization that build on the bound established here.

1 Introduction

The GFM framework defines its actor as a system that maximizes $\mathrm{vol_P}$—the poset measure over the capability space [Lasser, 2026b, Definition 6]. The axiomatization of [Lasser, 2026f, Proposition 1] establishes $\mathrm{vol_P}$ as a poset measure satisfying axioms M1–M6 and proves it self-balancing [Lasser, 2026f, Theorem 1]. The analysis of [Lasser, 2026c] extends to multi-substrate collectives, proving that full domination is anti-maximizing under discounting [Lasser, 2026c, Proposition 6]. Throughout, $\mathrm{vol}_{\mathrm{P}}$ is treated as a faithful proxy for the experiential optionality the framework is meant to protect.

But $\mathrm{vol}_{\mathrm{P}}$ measures what the collective *can* do, not what it *does*. The B-to-C gap—named in [Lasser, 2026b, Section 7] and flagged as a primary open problem in the sequence—is the possibility that these diverge. Two failure modes illustrate:

1. **The Doll Problem** [Lasser, 2026b, Section 7.1]: a collective possesses rich capabilities but exercises none. $\mathrm{vol}_{\mathrm{P}}$ is high but experiential value is zero.
2. **Wireheading** [Lasser, 2026a, Discussion]: a collective exercises capabilities in narrow self-reinforcing loops that serve the measurement system rather than experiential value. High $\mathrm{vol}_{\mathrm{P}}$ and apparent exercise, but the exercise is degenerate.

Both are instances of Goodhart's Law [Strathern, 1997, Manheim and Garrabrant, 2019]: when the measure ($\mathrm{vol}_{\mathrm{P}}$) diverges from the target (experiential optionality), optimizing the measure no longer serves the target.

1.1 Why direct measurement fails

A natural first approach is to define a *realized capability volume* $\mathrm{vol}_{\mathrm{R}}$ by restricting $\mathrm{vol}_{\mathrm{P}}$ to an exercised sub-poset—capabilities counterfactually load-bearing for realized outputs—and monitor the ratio $\beta = \mathrm{vol}_{\mathrm{R}}/\mathrm{vol}_{\mathrm{P}}$. This fails on two grounds:

Tractability failure. The counterfactual query is worst-case exponential in pathway length and requires global knowledge of every realized output the collective produces.

Privacy failure. Computing realized-output sets requires observational access to what the collective actually does. This is the total-surveillance scenario. The framework's operational commitments [Lasser, 2026a,e,h] rule out panopticon-style observation.

1.2 What this paper does

We observe that agents *voluntarily disclose* $\mathrm{vol}_{\mathrm{R}}$-content through an existing, privacy-respecting channel: **sacrifice**. When an agent surrenders a benchmarkable capability (whose $\mathrm{vol}_{\mathrm{P}}$ contribution is known) in exchange for an unbenchmarked bundle (whose $\mathrm{vol}_{\mathrm{R}}$ contribution is unknown), the rational-choice inequality reveals the unbenchmarked bundle is worth at least what was surrendered. Each such event produces a lower bound on a portion of $\mathrm{vol}_{\mathrm{R}}$. Aggregating across events with pairwise poset-independent bundles produces a constructive lower bound on the B-to-C gap.

This paper formalizes the sacrifice event (Section 2), states the six assumptions (S0–S5) under which the aggregate bound holds, proves the central theorem (Section 3), analyzes the two observation modalities (money and time, Section 4), establishes the structural non-self-balancing finding for $\mathrm{vol}_{\mathrm{R}}$ (Section 5), and composes the channel with the commitment protocol of [Lasser, 2026a] (Section 6).

1.3 Scope: what this paper does not do

- **It does not filter below-sufficiency sacrifices.** Not every observed sacrifice event is voluntary—agents pay below-sufficiency costs (dehydration, debt-servicing) not because they value the outcome highly but because they cannot opt out. Such events have reversed polarity relative to the theorem's assumptions (S1 fails). The companion paper [Lasser, 2026d] introduces the *need-sufficiency architecture* that filters these events structurally before they reach the observation channel. The present paper takes the filter as an external precondition: events reaching the aggregate are assumed above-sufficiency and S1-admissible.
- **It does not decompose the B-to-C gap into diagnostic categories.** Interpreting why $\beta^{\mathrm{lower}} = \mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}/\mathrm{vol}_{\mathrm{P}}$ is low requires partitioning capabilities into restricted, covered, dormant, and residual cells and tracking which cell dominates the gap. The gap decomposition machinery and its wireheading-consistent concentration signal are developed in [Lasser, 2026d].

- **It does not provide a complete theory of experiential value.** The residual class (capabilities structurally outside the sacrifice channel's reach) is characterized in [Lasser, 2026d].
- **It does not replace vol_P with vol_R.** The framework continues to optimize vol_P; the revealed-sacrifice lower bound is a diagnostic that detects when vol_P optimization has gone wrong. vol_R is the audit, not the objective.
- **It does not measure vol_R directly.** The paper produces a *lower bound* on vol_R from trade events, not a precise measurement.

1.4 Summary of results

1. **Privacy-minimal observation channel** (Section 2). Revealed-sacrifice observation defines the sacrifice event and states six assumptions (S0–S5). Three formal privacy properties are established (Theorem 1: no interior access, discretization, commitment-layer composability) and two institutional preconditions (Proposition 1: consent via participation, third-party observability) are documented as deployment-level dependencies.
2. **Aggregate B-to-C Lower Bound** (Section 3). Theorem 2 lower-bounds the exercised fraction β on the subset of capability-space covered by voluntary, above-sufficiency, assumption-admissible trade events with *pairwise poset-independent bundles*. Part A (bundle-level, requires S0–S2 and S4(a)) is the main result; Part B (per-capability, requires S0–S5 plus within-bundle poset-independence) is a conditional refinement. The constructed lower bound is non-decreasing in observed admissible events: Part A via sub-batch supremum (Proposition 3) and Part B via per-capability max-attribution (Proposition 2).
3. **Two sacrifice channels, substitutable with residuals** (Section 4). Money-sacrifice and time-sacrifice are substitutable via cooperative markets (employment converts time to money; hiring converts money to time). The channels triangulate the same underlying substrate rather than covering disjoint capability spaces. Non-fungibility residuals (agent-specific time, capital-concentrated access) are existential witnesses enabling strict improvement of the joint aggregate conditional on observed-coverage content (Corollary 1).
4. **Non-self-balancing finding** (Section 5). vol_R is non-invariant under capability removal: unexercised capabilities can be eliminated without reducing vol_R, while vol_P contracts by their weight. The self-balancing property therefore does not transfer from vol_P to vol_R, so vol_R is a diagnostic, not an objective. This structural finding is independent of the observation mechanism.
5. **Commitment-layer composition** (Section 6). The sacrifice channel composes with the commitment protocol of [Lasser, 2026a] to produce a privacy-minimal aggregate on distinct-category batches (Propositions 6 and 7). The rollup ZK-certifies the Part A formula inputs; the theorem's non-ZK hypotheses require external attestation (Remark 14).

1.5 Design principle

This paper shares a structural discipline with [Lasser, 2026e] and [Lasser, 2026h]: **resolution through frequency, not through depth.** Controlled relaxation [Lasser, 2026h] gains precision by running more tests, not by surveilling each test more intrusively. Revealed-sacrifice observation gains precision by observing more trades, not by observing each trader more intrusively.

1.6 Dependencies on prior papers

A table summarising the prior-paper results this paper relies on is collected in Appendix C.

2 Revealed-Sacrifice Observation Model

The privacy-compatible observation channel for vol_R is *revealed sacrifice*: agents disclose the value they place on unbenchmarked capabilities through the benchmarked capabilities they voluntarily surrender to obtain them. The framework does not observe *what* the agent values privately—only *what they commit to publicly through sacrifice*.

2.1 Core definitions

Definition 1 (Revealed-Sacrifice Event)**.** *A revealed-sacrifice event is a tuple (i, X, Y, t) where:*

- *i is the agent performing the sacrifice;*
- *X is a benchmarked capability (or capability bundle) with known $\Delta\mathrm{vol}_{\mathrm{P}}(X) \geq 0$, surrendered by agent i at time t;*
- *$Y = \{Y_1, \ldots, Y_k\}$ is an unbenchmarked capability bundle acquired by agent i, satisfying the* genuine- exchange conditions*: (i) $X \cap Y = \emptyset$ (the agent does not surrender and reacquire the same capability); (ii) $Y \cap (H_i \setminus X) = \emptyset$ (the acquired bundle is novel relative to the agent's retained holdings $H_i \setminus X$, where H_i is the agent's pre-trade holdings); (iii)* post-trade cooperative closure*: no cooperative capability in the poset has participants spanning both Y and $H_i \setminus X$ (so post-trade, no new cross-bundle cooperative capability is activated by the combination); and (iv)* pre-trade cooperative closure*: no cooperative capability in the poset has participants spanning both X and $H_i \setminus X$ (so pre-trade, no cooperative capability straddles the surrendered and retained capability sets—only cooperatives contained entirely in X or entirely in $H_i \setminus X$ exist);*
- *t is the event timestamp.*

Conditions (i) and (ii) characterize genuine *trades: the agent exchanges distinct capabilities and acquires something novel. Conditions (iii) and (iv) together ensure that the exchange-to-bundle cancellation in the revealed-preference derivation is exact: (iii) rules out post-trade cross-bundle cooperatives that would inflate $U_i((H \setminus X) \cup Y)$ beyond $U_i(H \setminus X) + U_i(Y)$; (iv) rules out pre-trade cross-bundle cooperatives that would inflate $U_i(H)$ beyond $U_i(H \setminus X) + U_i(X)$. Both closures are needed symmetrically for the cancellation of Lemma 1 to hold. Trades violating (i) or (ii) are degenerate; trades violating (iii) or (iv) carry revealed-preference information about the marginal contribution of Y in context rather than $\mathrm{vol}_{\mathrm{R}}(Y)$ in isolation—the bound remains conservatively valid for the contextual quantity but the theorem's isolated-$\mathrm{vol}_{\mathrm{R}}(Y)$ interpretation requires all four conditions. The event emits the signal:*

$$\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(Y) \;\geq\; \Delta\mathrm{vol}_{\mathrm{P}}(X)$$

under assumptions S0 (calibration), S1 (free choice), S2 (benchmark grounding), and S4(a) (valuation additivity) stated below. S4(a) is needed for the exchange-to-bundle cancellation (Lemma 1).

Lemma 1 (Exchange-to-Bundle Cancellation)**.** *Let H be a finite capability holding set with $X \subseteq H$, and let Y be a finite capability set satisfying the genuine-exchange conditions (i)–(iv) of Definition 1 relative to H: (i) $X \cap Y = \emptyset$; (ii) $Y \cap (H \setminus X) = \emptyset$; (iii) no cooperative capability in the poset has participants spanning both Y and $H \setminus X$; (iv) no cooperative capability in the poset has participants spanning both X and $H \setminus X$. Under S4(a) (global valuation additivity: $U_i(S) = \sum_{c \in S} u_i(c)$ for any finite $S \subseteq P$, with cooperative capabilities included as explicit nodes in S when their participants are all in S):*

$$U_i\big((H \setminus X) \cup Y\big) - U_i(H) \;=\; U_i(Y) - U_i(X).$$

Equivalently, $U_i((H \setminus X) \cup Y) \geq U_i(H) \iff U_i(Y) \geq U_i(X)$.

Proof. By S4(a), $U_i(A) = \sum_{c \in A} u_i(c)$ for any finite capability set A. To apply this cleanly we must track cooperative capabilities carefully: under S4(a), a cooperative capability c_{coop} contributes $u_i(c_{\mathrm{coop}})$ to $U_i(A)$ whenever $c_{\mathrm{coop}} \in A$, which holds iff all of c_{coop}'s participants are in A.

Pre-trade state H. $H = (H \setminus X) \sqcup X$. By (iv), no cooperative capability has participants spanning X and $H \setminus X$, so every cooperative capability in H has participants entirely in X or entirely in $H \setminus X$. Hence

$$U_i(H) = U_i(H \setminus X) + U_i(X),$$

where each term includes cooperatives formable within that set alone.

Post-trade state $(H \setminus X) \cup Y$. By (ii) this is a disjoint union: $(H \setminus X) \sqcup Y$. By (iii), no cooperative capability has participants spanning Y and $H \setminus X$, so every cooperative in the post-trade state has participants entirely in Y or entirely in $H \setminus X$. Hence

$$U_i\big((H \setminus X) \cup Y\big) = U_i(H \setminus X) + U_i(Y).$$

Cancellation. Subtracting, the $U_i(H \setminus X)$ terms cancel, leaving $U_i((H \setminus X) \cup Y) - U_i(H) = U_i(Y) - U_i(X)$.

Condition (i) makes $U_i(Y)$ and $U_i(X)$ refer to distinct capability sets; violating (i) would cause $X \cap Y$ terms to cancel trivially on both sides, yielding no information. ■

Justification (revealed-preference inequality). The transfer from the observed trade to the framework-level bound $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(Y) \geq \Delta\mathrm{vol}_{\mathrm{P}}(X)$ proceeds in three steps.

1. **Exchange preference (revealed-preference observation).** When agent i trades X for Y, revealed-preference theory [Samuelson, 1938] establishes that the post-trade state is weakly preferred to the pre-trade state: $U_i\big((H \setminus X) \cup Y\big) \geq U_i(H)$, where H is the agent's pre-trade holdings.
2. **Reduction to bundle-level comparison.** By Lemma 1 under S4(a) and the genuine-exchange conditions of Definition 1, the exchange comparison reduces to $U_i(Y) \geq U_i(X)$.
3. **Calibration chain.** Given $U_i(Y) \geq U_i(X)$, the bridge to the framework-level bound uses:
   - Sacrifice-side grounding (S2): $U_i(X) \geq \Delta\mathrm{vol}_{\mathrm{P}}(X)$, which holds by construction when $\mathrm{vol}_{\mathrm{P}}$ is the operational target on the benchmarked subspace.
   - Acquisition-side calibration (S0): $U_i(Y) \leq {\mathrm{vol}_{\mathrm{R}}}^{[W]}(Y)$, asserting that agents do not systematically overvalue unbenchmarked bundles relative to their *window-integrated* exercise contribution. The window-integrated form is what the observation channel actually bounds: an acquisition at t_n carries no instantaneous exercise witness, but the bound is still valid on the cumulative window quantity ${\mathrm{vol}_{\mathrm{R}}}^{[W]}$ whenever the agent uses Y during W.

   Chaining: ${\mathrm{vol}_{\mathrm{R}}}^{[W]}(Y) \geq U_i(Y) \geq U_i(X) \geq \Delta\mathrm{vol}_{\mathrm{P}}(X)$.

Remark 1 (S4(a) is a genuine assumption, not a consequence)**.** *S4(a) is a global additive-separability assumption on U_i. It is stronger than the rationality-only premises of Samuelson's original revealed-preference theorem [Samuelson, 1938], which establishes preferences over states rather than additivity over bundles. In applied microeconomics [Mas-Colell et al., 1995], bundle-level inference of the kind the sacrifice channel performs typically invokes either additive separability or quasilinearity in a numeraire. The paper uses a special case implicitly via the time-sacrifice channel (Definition 8), which values time at wage rate r_i—quasilinearity in wage-scaled time, a form of valuation additivity. S4(a) is the explicit assumption that makes Lemma 1 go through; it is not a consequence of S0–S3.*

Complementarities that would break pure additivity in general consumer theory are absorbed into the poset as cooperative nodes via M6 [Lasser, 2026f, Proposition 1]: cooperative capabilities are themselves capabilities with their own weights, so bundle-specific cooperative contributions appear as separate summands in U_i. S4(a)'s additivity is compatible with this structure because cooperative terms are explicit capabilities, not hidden non-linearities.

2.2 Assumptions

The revealed-sacrifice channel and its aggregate bound (Section 3) operate under six named assumptions. We collect them here for single-point reference.

S0 (Valuation-bounded-by-Exercise). For each agent i_n with event at $t_n \in W$, the agent's valuation of the unbenchmarked bundle Y_n does not exceed Y_n's *window-integrated* realized $\mathrm{vol}_{\mathrm{R}}$ contribution: $U_{i[n]}(Y_n) \leq {\mathrm{vol}_{\mathrm{R}}}^{[W]}(Y_n)$. This is the *substantive bridge* between valuation (what the agent pays for at t_n) and realized exercise (what the framework cares about, aggregated over the observation window W). It is not a calibration-in-the-narrow-sense (unbiased estimation) but a one-sided conservatism claim: agents do not systematically pay more than what the capability will actually deliver in exercise *within* W. The window-integrated form is essential: the sacrifice channel observes acquisition at t_n, not exercise at any particular time, so S0's consequent is about the cumulative exercise witness across the window rather than the instantaneous value at t_n. Agents who buy capabilities they never use within W violate S0 for those events; this matches the intuition that the lower bound is about *realized* rather than *stored* capability.

S1 (Free choice). Each sacrifice event is a voluntary trade: agent i_n chose Y_n over X_n when both were available. The agent had *non-degrading* alternatives: retaining X_n was a viable option that would not have left the agent below minimum viable functionality on any need dimension. When retaining X_n would cause the agent to fall below a sufficiency threshold, S1 fails for that event (see the companion paper [Lasser, 2026d] for the need-sufficiency architecture that filters such below-sufficiency events structurally).

S2 (Benchmark grounding). The benchmarked sacrifice X_n has a well-defined vol_P contribution $\Delta\mathrm{vol}_\mathrm{P}(X_n)$ computed via [Lasser, 2026f, Proposition 1], and the agent's valuation of X_n is at least as large: $U_i[_n](X_n) \geq \Delta\mathrm{vol}_\mathrm{P}(X_n)$.

S3 (Bundle coherence). The hedonic decomposition of each bundle Y_n into capabilities in U is well-defined (no multicollinearity degeneracy).

S4 (Additive separability).

(a) *Valuation additivity (global):* U_i is additive over any finite capability set: $U_i(S) = \sum_{c\in S} u_i(c)$ for any finite $S \subseteq P$, with $u_i(c) \geq 0$. In particular this applies to sacrifice bundles ($U_i(X) = \sum_{c\in X} u_i(c)$), acquired bundles ($U_i(Y)$), and agent holdings ($U_i(H)$). Complementarities are handled at the poset level by treating cooperative capabilities as their own nodes (M6), which appear as additional capabilities in S whose u_i accounts for the cooperative term.

(b) vol_R *additivity on poset-independent capabilities:* The vol_R contribution of a set of capabilities that are pairwise poset-independent (no subsumption relations *and* no cooperative composition links) is additive.

S5 (Decomposition validity). The event-local decomposition coefficients are *lower bounds* on the true vol_R share: $\alpha_{n,c} \leq \mathrm{vol}_\mathrm{R}(c)/\mathrm{vol}_\mathrm{R}(Y_n)$ for all n and $c \in Y_n$.

Remark 2 (S5 is a conditional-refinement assumption)**.** *S5 is the strongest of the six assumptions: it asserts a direct share inequality on the quantity Part B of Theorem 2 bounds. Part A makes no per-capability attribution claim and does not require S5. Part B's per-capability refinement should be understood as a* conditional result*: under the additional structural assumption that decomposition coefficients are conservative share lower bounds,* and *the hypothesis of within-bundle poset-independence (Definition 4, Remark 6), the aggregate can be refined to per-capability bounds. Within-bundle poset-independence is what makes S5's share inequality compatible with the partition-of-unity constraint on the α coefficients: bundles with internal cooperative activation produce* $\mathrm{vol}_\mathrm{R}(Y_n) > \sum_{c\in Y_n} \mathrm{vol}_\mathrm{R}(c)$*, forcing* $\sum_{c\in Y_n} \alpha_{n,c} < 1$ *under S5 and violating Definition 4. Operationally, S5 holds when decomposition coefficients are set via conservative heuristics (e.g., equal-weight default* $\alpha_{n,c} = 1/|Y_n|$ *when* Y_n *is a small homogeneous bundle, known-bias corrections from prior hedonic [Rosen, 1974] regressions) rather than via fitted models that could overstate individual capability contributions.*

Part A of Theorem 2 requires S0–S2 and S4(a): S4(a) is the valuation-additivity assumption used in the exchange-to-bundle reduction of the revealed- preference inequality (Step 2 of the justification above). Part B additionally requires S3 (bundle coherence), S4(b) (vol_R additivity on poset-independent capabilities), S5 (decomposition validity), and within-bundle poset-independence of each admissible Y_n (Definition 4). The need-sufficiency architecture of the next section characterizes when S1 fails structurally and provides the filter that keeps the sacrifice channel clean.

Remark 3 (S0 is the substantive bridge, not a calibration detail)**.** *S0 does the conceptual work of the theorem. Revealed-preference reasoning on its own establishes only that traded bundles are valued at least as highly as surrendered bundles; it does not establish that what the agent values corresponds to what the capability* delivers in exercise*. S0 asserts precisely that correspondence (in the one-sided direction needed for a lower bound): subjective valuation is upper-bounded by realized exercise contribution. This is a non-trivial behavioural claim. Agents who systematically overpay for status, novelty, or anticipated-but-unrealized utility violate S0 in realistic markets; the theorem's empirical validity depends on the prevalence of such violations.*

S0 does not *require that agent valuations perfectly track* vol_R*—only that they do not* overstate *it. Agents may undervalue unbenchmarked bundles (sacrifice little for something very valuable), in which case the lower bound is conservative but still valid. S0 fails only in the overpayment direction. Weakening S0 to expected/distributional conservatism (Open Question 6) is a priority direction for follow-up work.*

Definition 2 (Aggregate Trade Window)**.** *An aggregate trade window $[t_0, t_0 + T]$ is an observation period over which revealed-sacrifice events are collected. The window parameters are:*

- T*: window duration;*
- N*: number of observed sacrifice events in the window;*
- $\{(i_n, X_n, Y_n, t_n)\}_{n=1}^{N}$*: the event sequence.*

The aggregate trade window plays a role analogous to the observation windows used elsewhere in the sequence [Lasser, 2026g]: it defines the timescale over which the framework accumulates sacrifice evidence.

2.3 Privacy properties

The channel's privacy properties split cleanly into two categories: *formal* properties that follow directly from the observation-model construction (stated below as Theorem 1), and *institutional* preconditions that require deployment-level infrastructure to hold operationally (stated as Proposition 1). We state each category separately so that the theorem-level content covers only mathematical consequences of the construction, while institutional dependencies are flagged as preconditions rather than proved facts.

Theorem 1 (Formal Privacy Properties of Revealed-Sacrifice Observation)**.** *The revealed-sacrifice channel satisfies the following three formal privacy properties, each following from the observation tuple (Definition 1) together with the commitment protocol of [Lasser, 2026a, Definition 6]:*

(P1) ***No interior access.*** *The framework never observes what the agent values privately—only what they commit to publicly through sacrifice. The observation channel records the tuple (i, X, Y, t), not the agent's internal valuation function U_i.*

(P3) ***Discretization as a feature.*** *Trade events are sparse by construction. The channel's emission rate is bounded by trade frequency, not by observation intensity: each sacrifice event produces one output tuple, and no continuous stream is emitted between events. (This is an emission-rate bound, not a Shannon-information-rate bound; the information content per event is finite but non-trivial.)*

(P4) ***Commitment-layer composability.*** *The commitment protocol of [Lasser, 2026a, Definition 6] extends to revealed-sacrifice events: a sacrifice event can be committed as a proof-of-trade exposing only the $\mathrm{vol}_{\mathrm{R}}$-category and $\Delta\mathrm{vol}_{\mathrm{P}}$-magnitude, not counterparty, price, or specific goods (Section 6).*

Proof. Each property is established from the structure of the observation channel (Definition 1) and the aggregate bound (Theorem 2).

(P1) No interior access. The channel's output is, by Definition 1, the tuple (i, X, Y, t). The agent's internal valuation U_i appears only as a bridge quantity in the calibration chain (S0 + S1 + S2 + S4(a)); it is never recorded, transmitted, or stored by the framework. The aggregate bound (Theorem 2) depends only on $\{(\Delta\mathrm{vol}_{\mathrm{P}}(X_n), Y_n)\}$, so the framework's behaviour is identical under any two agent-valuation functions producing the same trade sequence. No interior access is required or obtained.

(P3) Discretization as a feature. The channel emits at most one output tuple per sacrifice event and produces no continuous stream otherwise. Over a window of duration T the total number of channel emissions is exactly the number of observed events, so the channel's *emission rate* is $|\mathrm{events}|/T$. Unlike continuous observation (which scales with sampling rate), the emission rate is bounded above by the agents' trade frequency—a behavioural rather than observational parameter—so increasing observation intensity does not increase the emission rate. (Each emission carries finite information about the tuple (i, X, Y, t); this is an emission-rate claim, not a Shannon-information-rate claim.)

(P4) Commitment-layer composability. Each sacrifice event admits a commitment $\Gamma = \mathrm{Commit}\big((\Delta\mathrm{vol}_{\mathrm{P}}(X), \mathrm{category}(Y)), r\big)$ using the protocol of [Lasser, 2026a, Definition 6], accompanied by a zero-knowledge proof [Lasser, 2026a, Definition 7] that Γ binds a valid revealed-sacrifice event: the event-local objectively verifiable facts—non-negativity of $\Delta\mathrm{vol}_{\mathrm{P}}(X_n)$, bundle

containment $Y_n \subseteq C_{\mathrm{category}(Y)[n]}$ (Definition 9(d)), bundle coherence S3, and a verifiable link to an underlying trade record. This is the event-level certification; S4(b) is not an event-level property but a batch-level structural fact about the category partition, established via pairwise-distinct category labels in Proposition 7 and inherited from the poset-independence of distinct partition components (Definition 10(i)). S0, S1, S4(a), and S5 are *not* ZK-verifiable (Remark 13). Proposition 6 establishes that the Part A aggregate bound is computable from committed events with *pairwise-distinct* category labels, together with the category-partition structure; batches with colliding labels require the operational responses of Remark 15 (sub-batch partitioning, finer public partition, or richer commitments). The committed form reveals only the sacrifice magnitude and bundle category; agent identity, specific X and Y, and counterparty remain hidden by the hiding property of Commit [Goldwasser et al., 1985]. ■

Proposition 1 (Institutional Preconditions for Revealed-Sacrifice Observation)**.** *Two additional privacy properties hold operationally when the deployment environment supplies the corresponding institutional infrastructure. These are not formal consequences of the channel's construction; they are preconditions the framework relies on but does not itself construct.*

(P2) Consent via participation (institutional). *The aggregate lower bound sums per-capability max-attribution quantities over observed events (Theorem 2, Part B), so an agent with zero observed sacrifice events contributes zero to the aggregate. This zero-contribution property is formal. The* operational *interpretation as "consent" additionally requires that participation in markets and labour exchanges be legally and socially opt-in; in settings where participation is coerced, the formal zero-contribution property holds but the consent interpretation does not.*

(P5) Third-party observability (institutional). Money-sacrifice events *are observable from public-facing ledger data (market exchanges, payment rails, public commitments) with no agent participation required,* assuming *such public-ledger infrastructure exists. The sacrifice magnitude $\Delta\mathrm{vol}_{\mathrm{P}}(X)$ is directly readable from the price paid.*

Time-sacrifice events *are observable from public-ledger data only under an additional precondition: the agent's outside-option wage rate r_i (Definition 8) must be computable from public inputs—e.g., a public wage schedule, occupational census with public imputation rules, or the agent's own published rate. Where r_i is private, the hours h may be observable but the sacrifice magnitude $\Delta\mathrm{vol}_{\mathrm{P}}(r_i \cdot h)$ is not third-party computable; P5 then holds for the money channel and for public-wage-imputation time events, but not for fully private time-sacrifice valuations.*

Where public-ledger infrastructure does not exist at all, P5 fails operationally on both channels even though the construction is unchanged. This contrasts with controlled relaxation [Lasser, 2026h] (framework must impose a test) and with commitment-protocol verification [Lasser, 2026a] (agent must generate a proof). P5 does not extend to private time-sacrifice activities (parenting, contemplation), which require voluntary disclosure regardless of ledger infrastructure and regardless of wage imputation.

Remark 4 (Why the split)**.** *Theorem 1 restricts itself to properties that are mathematical consequences of the channel's construction. Proposition 1 catalogues the deployment conditions under which P2 and P5 hold operationally. Treating P2 and P5 as institutional preconditions rather than as theorem consequences is the honest framing: they are load-bearing for how the channel behaves in practice, but they depend on facts about markets, legal systems, and public infrastructure that lie outside the paper's formal scope.*

3 Aggregate Lower Bound Theorem

3.1 Realized capability volume ($\mathrm{vol}_{\mathrm{R}}$)

Before stating the aggregate theorem, we make the object being bounded precise. $\mathrm{vol}_{\mathrm{R}}$ is the poset measure evaluated on the *exercised sub-poset*: the restriction of the capability space to capabilities that are currently being exercised.

Definition 3 (Exercised Sub-Poset and Realized Capability Volume)**.** *Let P be the full capability poset with measure $\mathrm{vol}_{\mathrm{P}}$ [Lasser, 2026f, Proposition 1].*

Instantaneous form. *For each time t, the* exercised sub-poset $P_t^{\text{ex}} \subseteq P$ *is the set of capabilities being exercised at time t—a* latent structural fact *about the system, independent of any observation mechanism:*

$$P_t^{\text{ex}} \;=\; \{d \in P : d \text{ is exercised at time } t\},$$

where "exercised at t" is formalized via the counterfactual exercise condition of Appendix A (Definition 11): d is exercised at t iff removing d at that time would make some realized cooperative output unrealizable. The instantaneous realized capability volume *is the poset measure restricted to this sub-poset:*

$$\mathrm{vol}_{\mathrm{R}t}(S) \;=\; \mathrm{vol}_{\mathrm{P}}\big(S \cap P_t^{\text{ex}}\big) \qquad \text{for any } S \subseteq P.$$

Window-integrated form. *For a trade window $W = [t_0, t_0 + T]$ (Definition 2), the* window-integrated exercised sub-poset *is the union of the instantaneous exercised sub-posets across the window,*

$$P_W^{\text{ex}} \;=\; \bigcup_{t \in W} P_t^{\text{ex}},$$

and the corresponding window-integrated realized capability volume *is*

$$\mathrm{vol_R}^{[W]}(S) \;=\; \mathrm{vol}_{\mathrm{P}}\big(S \cap P_W^{\text{ex}}\big) \qquad \text{for any } S \subseteq P.$$

$\mathrm{vol_R}^{[W]}$ *counts a capability as contributing to realized volume if it was exercised at* any *time during W; the instantaneous form $\mathrm{vol}_{\mathrm{R}t}$ counts only capabilities exercised exactly at t. Both inherit the measure structure of $\mathrm{vol_P}$ restricted to the corresponding sub-poset, and the axiom-analogue analysis of Section 5 applies to each.*

Convention. *Unless otherwise noted,* $\mathrm{vol_R}$ *without a time subscript refers to the window-integrated form* $\mathrm{vol_R}^{[W]}$ *for the ambient trade window W. This is the quantity the sacrifice channel (Theorem 2) actually bounds: a sacrifice event at t_n records an acquisition, not an instantaneous exercise witness, so the theorem's lower bound is stated against* $\mathrm{vol_R}^{[W]}$*—which captures any exercise of Y_n anywhere in W—rather than against the instantaneous* $\mathrm{vol}_{\mathrm{R}t}$ *at a particular evaluation time. Assumption S0 (Section 2.2) is correspondingly stated directly in window-integrated form: $U_i[n](Y_n) \le \mathrm{vol_R}^{[W]}(Y_n)$.*

Remark 5 ($\mathrm{vol_R}$ is latent; $\mathrm{vol}_{\mathrm{R}}^{\text{lower}}$ is the bound)**.** $\mathrm{vol_R}$ *(in either form) is defined structurally as a ground-truth quantity about the system; computing it* directly *requires evaluating the counterfactual exercise condition for every capability at every relevant time, which the Introduction rejects as intractable and privacy-incompatible at scale. The revealed-sacrifice observation channel does not observe P_t^{ex} or P_W^{ex} directly; it produces a lower bound* $\mathrm{vol}_{\mathrm{R}}^{\text{lower}}$ *on the window-integrated* $\mathrm{vol_R}^{[W]}$ *restricted to an observed sub-space $U \subseteq P$. Inside the observation perimeter O of Appendix A, direct evaluation of $e_t(d)$ is permitted (no privacy barrier within consented operational scope) and contributes to a combined diagnostic* $\mathrm{vol}_{\mathrm{R}}^{\text{combined}} = \mathrm{vol}_{\mathrm{R}}^{\text{exact}}(O) + \mathrm{vol}_{\mathrm{R}}^{\text{lower}}(P \setminus O)$*, which is a valid lower bound on* $\mathrm{vol_R}(P)$ *when O is boundary-closed from $P \setminus O$ in the full poset-independence sense (neither cooperative-composition nor subsumption links cross the boundary; see Definition 11 conditions BC1/BC2); a conservative lower bound otherwise (Appendix A, Remark 19).* $\mathrm{vol_R}$ *itself is the latent target;* $\mathrm{vol}_{\mathrm{R}}^{\text{lower}}$ *(and* $\mathrm{vol}_{\mathrm{R}}^{\text{exact}}$ *on O) are the observation-channel outputs the framework actually computes.*

3.2 Event-local decomposition

Definition 4 (Event-Local Bundle Decomposition)**.** *For each revealed-sacrifice event (i_n, X_n, Y_n, t_n) whose bundle Y_n is* within-bundle poset-independent *(no subsumption relations and no cooperative-composition links among the capabilities in Y_n; see Definition 10(i) applied internally to Y_n), define the* event-local *decomposition coefficients $\alpha_{n,c} \ge 0$ for each capability $c \in Y_n$, satisfying:*

$$\sum_{c \in Y_n} \alpha_{n,c} = 1 \qquad \text{(partition of unity within the bundle).}$$

The coefficients represent the share of bundle Y_n attributable to capability c, determined at the time of event n and fixed thereafter. *Each event carries its own decomposition, independent of other* *events. The decomposition can be obtained from prior hedonic regression [Rosen, 1974] on similar trades, domain knowledge, or the equal-weight default $\alpha_{n,c} = 1/|Y_n|$.*

Within-bundle poset-independence is the hypothesis that makes the partition of unity compatible with S5's share inequality. Under within-bundle independence, no cooperative node activates from within Y_n, so $\mathrm{vol}_\mathrm{R}(Y_n) = \sum_{c\in Y_n} \mathrm{vol}_\mathrm{R}(c)$ by axiom M4, and S5's $\alpha_{n,c} \le \mathrm{vol}_\mathrm{R}(c)/\mathrm{vol}_\mathrm{R}(Y_n)$ sums to at most one across $c \in Y_n$ with equality achievable. Bundles with internal cooperative activation fall back to Part A only (see Remark 6).

Remark 6 (Why within-bundle independence is a Part B hypothesis)**.** *If Y_n contains capabilities a, b whose cooperative composition z has all participants in Y_n, then S4(a) (Section 2.2) treats z as an explicit bundle summand, so $\mathrm{vol}_\mathrm{R}(Y_n) = \mathrm{vol}_\mathrm{R}(a) + \mathrm{vol}_\mathrm{R}(b) + \mathrm{vol}_\mathrm{R}(z) > \sum_{c\in Y_n} \mathrm{vol}_\mathrm{R}(c)$. S5's share inequality $\alpha_{n,c} \le \mathrm{vol}_\mathrm{R}(c)/\mathrm{vol}_\mathrm{R}(Y_n)$ is then forced below $\mathrm{vol}_\mathrm{R}(c)/(\mathrm{vol}_\mathrm{R}(a) + \mathrm{vol}_\mathrm{R}(b) + \mathrm{vol}_\mathrm{R}(z))$ for each atomic $c \in Y_n$, and $\sum_{c\in Y_n} \alpha_{n,c} \le [\mathrm{vol}_\mathrm{R}(a) + \mathrm{vol}_\mathrm{R}(b)]/\mathrm{vol}_\mathrm{R}(Y_n) < 1$, contradicting the partition of unity. Part B's per-capability attribution therefore requires within-bundle poset-independence. Part A places no such constraint: its bundle-level bound $\mathrm{vol}_\mathrm{R}(Y_n) \ge \Delta\mathrm{vol}_\mathrm{P}(X_n)$ is valid regardless of internal cooperative structure.*

Remark 7 (Event-local versus global regression)**.** *Classical hedonic regression [Rosen, 1974] fits a* global *model across all observed trades simultaneously, producing the tightest decomposition but breaking the monotonicity of the max-attribution bound (Proposition 2) when new observations refit all coefficients. The event-local formulation fixes coefficients at observation time, sacrificing fit quality for the monotonicity guarantee.*

3.3 Central result

Theorem 2 (Aggregate B-to-C Lower Bound)**.** *Fix a trade window $W = [t_0, t_0 + T]$ (Definition 2) and a sequence of revealed-sacrifice events $\{(i_n, X_n, Y_n, t_n)\}_{n=1}^N$ with $t_n \in W$, whose bundles jointly cover a subset U of unbenchmarked capability-space. All vol_R quantities below refer to the* window-integrated *form ${\mathrm{vol}_\mathrm{R}}^{[W]}$ (Definition 3): the theorem bounds the realized volume accumulated during W, not the instantaneous $\mathrm{vol}_{\mathrm{R}t}$ at any single evaluation time.*

Part A (Main result — bundle-level lower bound; requires S0–S2 and S4(a)). *For each event n:*

$$ {\mathrm{vol}_\mathrm{R}}^{[W]}(Y_n) \;\ge\; \Delta\mathrm{vol}_\mathrm{P}(X_n). $$

For N events with pairwise poset-independent *bundles:*

$$ {\mathrm{vol}_\mathrm{R}}^{U,[W]} \;\ge\; \sum_{n=1}^{N} \Delta\mathrm{vol}_\mathrm{P}(X_n), $$

where ${\mathrm{vol}_\mathrm{R}}^{U,[W]}$ is the window-integrated realized capability volume restricted to the unbenchmarked subset U covered by the union of bundles. Part A is the paper's substantive result: the vol_P-to-vol_R lower bound per unit of observed sacrifice magnitude.

Part B (Conditional refinement — per-capability attribution; requires S0–S5 and within-bundle poset-independence). *Restrict attention to events n whose bundles Y_n are within-bundle poset-independent (Definition 4). When bundles overlap and per-capability attribution is desired, define for each capability $c \in U$:*

$$ \mathrm{vol}_\mathrm{R}^{\mathrm{lower}}(c) \;=\; \max_{n:\, c\in Y_n} \alpha_{n,c} \cdot \Delta\mathrm{vol}_\mathrm{P}(X_n). $$

Then, provided the capabilities $c \in U$ are pairwise poset-independent:

$$ {\mathrm{vol}_\mathrm{R}}^{U,[W]} \;\ge\; \sum_{c\in U} \mathrm{vol}_\mathrm{R}^{\mathrm{lower}}(c). $$

Part B is conditional on S5*—a direct share inequality on the quantity being bounded (see Remark 2)—and should be read as a refinement when admissible, not as a result comparable in strength to Part A.*

Proof. Part A. Fix an event n with $t_n \in W$. By S1, agent i_n chose Y_n over X_n while both were available. Revealed preference gives $U_i[n]\big((H_n \setminus X_n) \cup Y_n\big) \geq U_i[n](H_n)$ for pre-trade holdings H_n. By Lemma 1 under S4(a) and the genuine- exchange conditions of Definition 1, this reduces to $U_i[n](Y_n) \geq U_i[n](X_n)$. By S0 (valuation-bounded-by-exercise, stated in window-integrated form), $U_i[n](Y_n) \leq \mathrm{vol_R}^{[W]}(Y_n)$. By S2 (benchmark grounding), $U_i[n](X_n) \geq \Delta\mathrm{vol_P}(X_n)$. Chaining:

$$\mathrm{vol_R}^{[W]}(Y_n) \;\geq\; U_i[n](Y_n) \;\geq\; U_i[n](X_n) \;\geq\; \Delta\mathrm{vol_P}(X_n),$$

which is the per-event bound. The window-integrated form is essential: the sacrifice channel observes acquisition of Y_n at t_n, not exercise at any particular instant. S0's consequent is therefore stated as a bound on $\mathrm{vol_R}^{[W]}(Y_n)$—the cumulative exercise witness for Y_n across W—not on the instantaneous $\mathrm{vol}_{\mathrm{R}t}(Y_n)$ at any single evaluation time t, which could be zero even when Y_n was exercised earlier or later in W.

Suppose the N bundles $\{Y_n\}$ are pairwise poset-independent: no subsumption relation and no cooperative-composition link connects any capability in Y_n to any capability in $Y_{n'}$ for $n \neq n'$. Then the sub-posets carried by the bundles are disconnected components of the sub-poset on $U = \bigcup_n Y_n$. Axiom M4 of [Lasser, 2026f, Proposition 1] makes the poset measure additive over poset-disjoint components; $\mathrm{vol_R}^{[W]}$ inherits M4 (Section 5, Theorem 3), so $\mathrm{vol_R}^{[W]}(U) = \sum_n \mathrm{vol_R}^{[W]}(Y_n)$. Summing the per-event bounds:

$$\mathrm{vol_R}^{U,[W]} \;=\; \sum_{n=1}^{N} \mathrm{vol_R}^{[W]}(Y_n) \;\geq\; \sum_{n=1}^{N} \Delta\mathrm{vol_P}(X_n),$$

establishing Part A.

Part B. Part B is stated for events with $\Delta\mathrm{vol_P}(X_n) > 0$ and bundles Y_n that are within-bundle poset-independent (Definition 4); events with zero sacrifice carry no information, and bundles with internal cooperative activation fall back to Part A (Remark 6). Fix an admissible event n and its bundle Y_n. By Part A, $\mathrm{vol_R}(Y_n) \geq \Delta\mathrm{vol_P}(X_n) > 0$, so $\mathrm{vol_R}(Y_n) > 0$ and division is well-defined. S3 (bundle coherence) additionally ensures the hedonic decomposition of Y_n is non-degenerate, so $\alpha_{n,c}$ is well-specified.

Within-bundle poset-independence plus M4 (Section 5, Theorem 3) gives $\mathrm{vol_R}(Y_n) = \sum_{c \in Y_n} \mathrm{vol_R}(c)$: no cooperative node activates from within Y_n, so the bundle's $\mathrm{vol_R}$ decomposes atomically. S5 then asserts that the event-local coefficients are lower bounds on the true atomic shares:

$$\alpha_{n,c} \;\leq\; \mathrm{vol_R}(c)/\mathrm{vol_R}(Y_n), \qquad c \in Y_n,$$

which is feasible with $\sum_{c \in Y_n} \alpha_{n,c} = 1$ precisely under within-bundle independence. Rearranging and applying the Part A per-event bound:

$$\mathrm{vol_R}(c) \;\geq\; \alpha_{n,c} \cdot \mathrm{vol_R}(Y_n) \;\geq\; \alpha_{n,c} \cdot \Delta\mathrm{vol_P}(X_n),$$

for every $c \in Y_n$. If c appears in multiple admissible bundles, each quantity $\alpha_{n,c} \cdot \Delta\mathrm{vol_P}(X_n)$ is a valid lower bound on $\mathrm{vol_R}(c)$; taking the maximum preserves validity:

$$\mathrm{vol_R^{lower}}(c) \;:=\; \max_{n:\, c \in Y_n} \alpha_{n,c} \cdot \Delta\mathrm{vol_P}(X_n) \;\leq\; \mathrm{vol_R}(c).$$

(Using the max rather than a sum is what underwrites the monotonicity of Proposition 2: additional events can only weakly increase a max.)

By the Part B hypothesis, the capabilities $c \in U$ are pairwise poset-independent, so a second application of M4 (to the poset of capabilities, not bundles) gives global additivity:

$$\mathrm{vol_R}^{U} \;=\; \sum_{c \in U} \mathrm{vol_R}(c) \;\geq\; \sum_{c \in U} \mathrm{vol_R^{lower}}(c).$$

∎

Remark 8 (Why poset-independence, not set-disjointness)**.** *Set-disjointness ($Y_1 \cap Y_2 = \emptyset$) ensures no* capability *appears in two bundles, but does not rule out subsumption or cooperative links across bundles. If a cooperative capability c_{coop} depends on capabilities in both Y_1 and Y_2, its* $\mathrm{vol_R}$ *contribution is not attributable to either bundle alone. Poset-independence rules this out: under poset-independence, the sub-posets are disconnected components and* $\mathrm{vol_P}$ *(hence* $\mathrm{vol_R}$*) decomposes additively by axiom M4 [Lasser, 2026f, Proposition 1].*

Definition 5 (Subset and Category-Level $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$)**.** *Under the Part B hypotheses (S0–S5) of Theorem 2, the per-capability lower bound* $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(c)$ *extends naturally to subsets of capability-space. For any* $A \subseteq P$*:*

$$\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(A) \;=\; \sum_{c \in A \cap U} \mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(c),$$

where U *is the observed subset covered by bundles. In particular:*

- *For a bundle category* $C_j \in \mathcal{C}$ *(Definition 10):* $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(C_j) = \sum_{c \in C_j \cap U} \mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(c)$.
- *For the restriction set* Ξ*, residual class* R_{S}*, and other capability subsets used in the companion paper's gap decomposition [Lasser, 2026d]:* $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(A)$ *is defined analogously.*

This extension is meaningful only under Part B. Under Part A alone, only the per-bundle quantities $\mathrm{vol}_{\mathrm{R}}(Y_n) \geq \Delta\mathrm{vol}_{\mathrm{P}}(X_n)$ *and the global aggregate* $\mathrm{vol}_{\mathrm{R}}{}^{U} \geq \sum_n \Delta\mathrm{vol}_{\mathrm{P}}(X_n)$ *are available. The gap-decomposition and wireheading-detection machinery of [Lasser, 2026d] is stated under Part B semantics unless otherwise noted.*

3.4 B-to-C ratio and monotonicity

Definition 6 (B-to-C Ratio Under Revealed Sacrifice)**.** *The B-to-C ratio under revealed-sacrifice observation is:*

$$\beta^{\mathrm{lower}}(G, T) \;=\; \mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(G, T) \;/\; \mathrm{vol}_{\mathrm{P}}(G),$$

where $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$ *is the aggregate lower bound from Theorem 2 and* T *indexes the trade window.* $\beta^{\mathrm{lower}} \in [0,1]$ *because* $\mathrm{vol}_{\mathrm{R}} \leq \mathrm{vol}_{\mathrm{P}}$ *by definition (the exercised sub-poset is a subset of the full poset).*

Proposition 2 (Part B Monotone Accumulation Under Max-Attribution)**.** *Under Part B hypotheses (S0–S5 and within-bundle poset-independence), define the per-capability lower bound as the maximum attribution across all observed events:*

$$\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(c) \;=\; \max_{n:\, c \in Y_n} \alpha_{n,c} \cdot \Delta\mathrm{vol}_{\mathrm{P}}(X_n),$$

and the aggregate as $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}} = \sum_{c \in U} \mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(c)$*. Then the aggregate is non-decreasing in the number of observed admissible events: for event sets* $E \subseteq E'$ *both satisfying the Part B admissibility conditions,*

$$\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(E') \;\geq\; \mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(E).$$

Proof. Fix $c \in U$. Let $S_N(c) = \{n \leq N : c \in Y_n\}$ be the set of events in the first N observations that cover c. For $N' > N$, $S_{N'}(c) \supseteq S_N(c)$, and

$$\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}\,(N')}(c) = \max_{n \in S_{N'}(c)} \alpha_{n,c}\Delta\mathrm{vol}_{\mathrm{P}}(X_n) \;\geq\; \max_{n \in S_N(c)} \alpha_{n,c}\Delta\mathrm{vol}_{\mathrm{P}}(X_n) = \mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}\,(N)}(c),$$

because a maximum over a superset is weakly larger. If $c \notin Y_n$ for any n then both sides are zero by convention and the inequality is trivial. Summing the per-capability inequalities over $c \in U$: $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}\,(N')} \geq \mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}\,(N)}$. ∎

Proposition 3 (Part A Monotone Accumulation via Sub-Batch Supremum)**.** *Under Part A hypotheses (S0–S2, S4(a), and the genuine-exchange conditions of Definition 1), define the Part A lower bound as the supremum over admissible pairwise-poset-independent sub-batches of the observed event set:*

$$\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}\,\mathrm{A}}(E) \;:=\; \sup\Big\{ \sum_{n \in B} \Delta\mathrm{vol}_{\mathrm{P}}(X_n) \;:\; B \subseteq E \text{ and } B \text{ satisfies the Part A admissibility and pairwise poset-independence}$$

with the supremum attaining 0 *on the empty sub-batch. Then* $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}\,\mathrm{A}}$ *is non-decreasing in* E*: for event sets* $E \subseteq E'$*,*

$$\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}\,\mathrm{A}}(E') \;\geq\; \mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}\,\mathrm{A}}(E).$$

Proof. Any admissible sub-batch $B \subseteq E$ is also an admissible sub-batch $B \subseteq E'$ (admissibility is a property of B itself, not of its ambient event set). Hence the supremum over admissible sub-batches of E is a supremum over a subset of the admissible sub-batches of E', and the supremum over a superset is weakly larger. ∎

Remark 9 (Interpretation of Part A monotonicity)**.** $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower\,A}}(E)$ *is the tightest Part A lower bound the framework can extract from the event set E by choosing which sub-batches to apply Theorem 2 to. Operationally, this is "apply the theorem to each maximal admissible sub-batch and take the sum (over poset-disjoint components of the covered space, with max-attribution or union-bookkeeping where batches overlap)." The sup-over- sub-batches definition is the principled monotone functional corresponding to the Part A bound;* $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower\,A}}$ *is what the paper refers to when it says "the constructed Part A lower bound is non-decreasing in observed admissible events."*

4 Two Sacrifice Channels: Money and Time

The sacrifice dimension splits into two observation channels. Money-sacrifice and time-sacrifice are not disjoint coverage spaces; they are *different modalities of the same underlying sacrifice substrate*, connected by cooperative-market exchange. The value of observing both is triangulation and robustness, with genuine strict inequality arising only at specific non-fungibility residuals.

4.1 Money-sacrifice channel

Definition 7 (Money-Sacrifice Event)**.** *A money-sacrifice event is a revealed-sacrifice event (i, X, Y, t) where the benchmarked sacrifice X is purchasing power: agent i pays a price p for an unbenchmarked bundle Y. The signal is:*

$$\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(Y) \;\geq\; \Delta\mathrm{vol}_{\mathrm{P}}(p),$$

where $\Delta\mathrm{vol}_{\mathrm{P}}(p)$ is the $\mathrm{vol}_{\mathrm{P}}$ contribution of the purchasing power surrendered.

Properties of the money channel:

- **Signal strength.** Price is a precise, quantitative signal. Market aggregation handles distributional loading (many agents trading the same category of goods produces a robust estimate).
- **Bias.** Privileges capabilities valued by wealthier agents. An agent with more purchasing power emits stronger signals per trade.
- **Third-party observability.** Money-sacrifice events leave public ledger traces (prices, timestamps, category labels); auditors can compute the money-channel aggregate without agent cooperation (Property P5 of Proposition 1).

4.2 Time-sacrifice channel

Definition 8 (Time-Sacrifice Event)**.** *A time-sacrifice event is a revealed-sacrifice event (i, X, Y, t) where the benchmarked sacrifice X is the agent's outside-option labour hours, valued at market wage rate r_i. Agent i foregoes $r_i \cdot h$ hours of wage income to engage in an unbenchmarked activity (volunteering, parenting, craft practice, relationship-building, contemplation). The signal is:*

$$\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(Y) \;\geq\; \Delta\mathrm{vol}_{\mathrm{P}}(r_i \cdot h).$$

Properties of the time channel:

- **Signal strength.** Opportunity cost is a quantitative signal grounded in market wage rates. Every agent has a time endowment and an outside option, regardless of wealth.
- **Bias.** More egalitarian than the money channel. Time is the scarcest capability every agent shares. A high-wage agent's time sacrifice has higher $\Delta\mathrm{vol}_{\mathrm{P}}$ per hour, but every agent sacrifices time.

- **Non-market agents.** Agents not in the labour market (retirees, children, homemakers, the unemployed) have an undefined or zero market wage rate. The framework can use imputed wage rates [Aguiar and Hurst, 2007] or non-market time valuation (replacement cost) to extend coverage to these populations.
- **Public versus private sacrifice.** Publicly observable time-sacrifice (committed volunteer hours, time-stamped artefact production) is third-party observable. Private time-sacrifice (parenting, contemplation) is visible only via voluntary disclosure—the residual flagged in Property P5 of Proposition 1.

4.3 Substitutability via cooperative markets

Money and time are not disjoint sacrifice types. Cooperative markets exchange between them:

- **Time $\to$ money.** Employment converts labour hours to wage income at rate r_i. This is the definitional exchange rate already embedded in Definition 8.
- **Money $\to$ time.** Agents can hire others to invest labour hours (services, delegation) or purchase time-saving technology and infrastructure (automation, efficiency-enhancing tools). The exchange goes through the same cooperative market that sustains employment.

Because the two channels are linked by this exchange, an agent's total sacrifice budget can be measured in a common monetary unit via the wage rate, and the agent can shift sacrifice between modalities. A high-earning agent can substitute money for time by hiring help; a time-rich agent can substitute time for money by investing hours directly. Neither channel *dominates* the other—each is the natural view of the same underlying sacrifice from one side of the exchange.

The practical consequence: **the two channels triangulate the same capability space, they do not partition disjoint capability spaces**. Observing both yields:

- **Triangulation.** When money-sacrifice and time-sacrifice both indicate high $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$ on a bundle, the signals corroborate. Divergence between the channels is itself diagnostic: a capability with strong time-sacrifice evidence but no money-sacrifice evidence in a given window may indicate the capability is exercised by time-rich agents outside the monetary economy, or that market mechanisms have not formed around it.
- **Robustness.** Different agent populations emit different modal signals. Children, retirees, and non-market agents emit time signals with imputed wage rates; wage- constrained high earners emit money signals. Observing both modalities extends the bound across populations that either alone would miss.

4.4 Non-fungibility residuals

The substitutability in §4.3 is not total. Two specific residuals create genuine asymmetries where one channel captures something the other cannot substitute for.

Proposition 4 (Agent-Specific-Time Residual (existential))**.** *There exists a capability c^* reachable by the time channel but whose $\mathrm{vol}_{\mathrm{R}}$-content is not captured by any money-channel event on c^* itself (access goods purchasable with money attach $\mathrm{vol}_{\mathrm{R}}$-content to* access categories*, not to c^*). The non-substitutability is of the specific c^*, not a claim that all agent-specific-time capabilities evade the money channel entirely.*

Proof. We exhibit $c^* =$ "sustained mentoring relationship with a specific individual." The relational capability is constituted by the interaction between this agent and the mentee; it is not transferable to a proxy. Money can purchase access goods (introductions, a mentor's scheduled time, coaching infrastructure), but these are distinct bundles whose $\mathrm{vol}_{\mathrm{R}}$-content lies in access categories rather than in the relational capability c^* itself. A paid mentoring engagement generates a money-channel event on *scheduled access* rather than on the relationship capability c^*. When agent i allocates h hours foregoing wage income $r_i \cdot h$ to the relationship, the time-sacrifice event (Definition 8) produces $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(Y) \geq \Delta\mathrm{vol}_{\mathrm{P}}(r_i \cdot h)$ with $c^* \in Y$, giving the time channel an event that the money channel does not produce for c^*. ∎

Proposition 5 (Capital-Concentrated Residual (existential)). *There exists a capability c^* reachable by the money channel but whose vol_R-content is not captured by any time-channel event on c^* itself (direct time-sacrifice on c^* without capital access yields no realized exercise). The non-substitutability is of the specific c^*, not a claim that all capital-concentrated capabilities evade the time channel entirely.*

Proof. We exhibit $c^* =$ "conduct a particle-physics experiment using a synchrotron." The exercise of c^* requires access to specialized infrastructure whose acquisition or lease cost is concentrated in a single transaction (institutional budget, grant award, facility time purchased). An agent accumulating time-sacrifice at wage rate r_i can only reach this capability by first converting time to money via employment and then purchasing access—which is the money channel, not the time channel, since the converted-to-money event is a market transaction. Direct time-sacrifice (hours spent without capital access) yields no experimental event and no vol_R signal on c^*. The money channel captures c^* via the single access-purchase event; the time channel is silent for c^* because the time-sacrifice exchange rate on direct-labor hours is too small relative to the capital threshold, and time-channel events accompanying capital-enabled exercise are attributable to labor categories rather than to c^*'s capital-concentrated access. ■

Remark 10 (Scope of the residual propositions). *Propositions 4 and 5 are* existential*—each witnesses one capability exhibiting the claimed residual behavior. They do not establish that the descriptive class "all agent-specific-time capabilities" or "all capital-concentrated capabilities" is uniformly channel-exclusive. Mixed-mode coverage of such capabilities can and does occur (a paid mentor also incurs time-sacrifice; a well-funded scientist can generate labor-attributed time events alongside capital-access events). The existential residuals are sufficient for Corollary 1's strict-improvement claim, which is stated conditional on the trade window actually containing at least one residual-category event.*

Corollary 1 (Joint Channel Triangulation). *Let E_{money} and E_{time} denote the money-channel and time-channel event sets over a trade window W, and suppose the union $E = E_{\mathrm{money}} \cup E_{\mathrm{time}}$ satisfies the closure conditions of Theorem 2 Part A: the bundles $\{Y_n\}_{n \in E}$ are pairwise poset-independent across channels—no subsumption relation and no cooperative-composition link connects any capability in a money-channel bundle to any capability in a time-channel bundle (set-disjointness of bundles across channels is necessary but not sufficient).*

Weak containment (unconditional). *The joint aggregate captures* weakly more *of U than either channel alone: any event observed through either channel alone is in the union, and Part A aggregation is monotone in observed admissible events (Proposition 2).*

Strict improvement (conditional on observed coverage). *The non-fungibility residuals (Propositions 4 and 5) make strict improvement* possible*: agent-specific-time capabilities lie in the coverage gap of the money channel alone, and capital-concentrated capabilities lie in the coverage gap of the time channel alone. Strictness over the money channel alone obtains when W contains at least one admissible time-only residual event; strictness over the time channel alone obtains when W contains at least one admissible money-only residual event. The residuals establish the* availability *of strict improvement, not its automatic realization; whether any given window realizes strictness is a data-coverage question.*

Proof sketch. Weak containment under the stated closure hypothesis: any event observed through either channel alone is also observed through the union, and the cross-channel pairwise poset-independence ensures Part A's aggregation is well-defined on E; the Part A lower bound is monotone in observed admissible events under sub-batch supremum (Proposition 3), so the union's bound is $\geq$ either single channel's bound. Strict improvement on the residuals: Proposition 4 witnesses a capability captured by time but not by money; Proposition 5 witnesses a capability captured by money but not by time. Both residuals guarantee strict improvement of the joint aggregate over either channel alone when the trade window includes at least one event from the relevant residual category. Outside the residuals, strictness becomes a data-coverage statement: the joint aggregate strictly exceeds either single channel if and only if the trade window includes events in categories the other channel does not cover. Substitutability in principle does not guarantee equality in practice; heterogeneous agent populations generically produce strict improvement, but a theoretical population using one channel exclusively could produce single-channel coverage equal to the joint aggregate. ■

Remark 11 (Cross-channel poset coupling)**.** *The closure hypothesis rules out all three forms of cross-channel coupling: (a) a capability c appearing in both a money-channel bundle and a time-channel bundle (set-overlap), (b) subsumption across channels ($c \succeq c'$ with c in one channel's bundle and c' in the other's), and (c) cooperative composition linking capabilities across channels. When any of these occurs in practice, the operational responses of Remark 15 apply: partition events into maximal poset-independent sub-batches per channel and aggregate separately, or fall back to Part B's per-capability max-attribution (which natively handles overlapping events via the max across events covering each capability, subject to Part B's additional hypotheses).*

4.5 The residual intersection

What neither channel catches is the *residual class* (characterized in the companion paper [Lasser, 2026d]): capabilities exercised purely privately, with no trade event, no publicly-observable time expenditure, and no sacrifice footprint of any kind. The residual class is the intersection of the coverage gaps of both channels after accounting for the cooperative-market substitutability of §4.3 and the non-fungibility residuals of Propositions 4 and 5. This is a strictly smaller set than the union of the two channels' individual coverage gaps: capabilities in one channel's gap but reachable via the other (or via the substitution markets) are not residual.

5 Axiom Inheritance and the Non-Self-Balancing Finding

Since $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$ is a lower bound on $\mathrm{vol}_{\mathrm{R}}$, and $\mathrm{vol}_{\mathrm{R}} = \mathrm{vol}_{\mathrm{P}}$ restricted to the exercised sub-poset, the axiom inheritance analysis applies directly, with an additional layer: the lower bound itself is not a measure (it is a bound on a measure).

Theorem 3 (Axiom-Analogue Analysis for $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$)**.** $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$ *is a lower-bound statistic, not a poset measure in the sense of [Lasser, 2026f, Proposition 1]; the clauses below describe* operational analogues *of M1–M6 for* $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$*, not literal axiom inheritance. M4 and the structural M6-failure result are substantive; M1, M2, M3, and M5 are weaker conditional statements noted for completeness.*

(M1) Non-negativity (analogue). *Satisfied.* $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$ *is a sum of non-negative terms ($\Delta\mathrm{vol}_{\mathrm{P}}(X_n) \geq 0$ per event; max-attribution preserves non-negativity).*

(M2) Null empty set (analogue). *Satisfied. With zero sacrifice events,* $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}} = 0$*.*

(M3) Monotonicity (conditional analogue). *Satisfied in the observational direction: adding sacrifice events covering a larger subset of U weakly increases* $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$ *(Proposition 2). M3 does* not *hold for the relationship between* $\mathrm{vol}_{\mathrm{P}}$ *and* $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$*: adding a benchmarked capability to the poset increases* $\mathrm{vol}_{\mathrm{P}}$ *but does not affect* $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$ *unless that capability is subsequently sacrificed. This is a lower-bound statistic property, not a poset-measure axiom.*

(M4) Additivity under disjointness (substantive). *Satisfied. By construction,* $\mathrm{vol}_{\mathrm{R}}$ *is* $\mathrm{vol}_{\mathrm{P}}$ *restricted to the exercised sub-poset $P^{\mathrm{ex}} \subseteq P$. M4 for* $\mathrm{vol}_{\mathrm{P}}$ *on P states that* $\mathrm{vol}_{\mathrm{P}}$ *is additive over components of P that are pairwise poset-disjoint (no subsumption or cooperative-composition links). Let $A, B \subseteq P^{\mathrm{ex}}$ be poset-disjoint in P^{ex}. Since P^{ex} inherits the relation structure of P, A and B are also poset-disjoint in P. M4 on P gives* $\mathrm{vol}_{\mathrm{P}}(A \cup B) = \mathrm{vol}_{\mathrm{P}}(A) + \mathrm{vol}_{\mathrm{P}}(B)$*; restricting both sides to P^{ex} gives* $\mathrm{vol}_{\mathrm{R}}(A \cup B) = \mathrm{vol}_{\mathrm{R}}(A) + \mathrm{vol}_{\mathrm{R}}(B)$*, establishing M4 for* $\mathrm{vol}_{\mathrm{R}}$*. This inheritance is load-bearing for the aggregate bound (Theorem 2, Part A).*

(M5) Non-triviality (conditional analogue). *A single sacrifice event with* $\Delta\mathrm{vol}_{\mathrm{P}}(X) > 0$ *produces* $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}} > 0$ *on the capabilities the event covers. This is an observation-dependent analogue, not the structural non-triviality of every valid capability.*

(M6) Superadditivity under independence. *The situation splits into a measure-axiom layer and a system-level layer:*

(a) $\mathrm{vol}_{\mathrm{R}}$ ***as a measure on a fixed exercised sub-poset inherits M6.*** *For any fixed P_W^{ex}, if $A, B \subseteq P$ are poset-independent in P, then $A \cap P_W^{\mathrm{ex}}$ and $B \cap P_W^{\mathrm{ex}}$ are poset-*

independent in P_W^{ex}, and $\mathrm{vol_P}$'s M6 on P lifts to $\mathrm{vol_R}$'s M6 on P_W^{ex} by the same restriction argument used for M4 above. This is literal axiom inheritance for the set function $\mathrm{vol_R}$.

The structurally relevant failure is at the system level*, not the measure-axiom level:* $\mathrm{vol_R}$ is non-invariant under capability removal. *Removing an unexercised capability c (a capability with $e_t(c) = 0$ throughout W) from the system leaves* $\mathrm{vol_R}$ *unchanged while* $\mathrm{vol_P}$ *drops by $w(c) > 0$. Example 1 witnesses this. The property is about the mapping $\mathcal{P} \mapsto \mathrm{vol_R}^{\mathcal{P}}$ from capability systems to the measure, not about* $\mathrm{vol_R}$ *as a set function on any one P_W^{ex}; it is what breaks the self-balancing transfer (Corollary 2).*

(b) $\mathrm{vol}_\mathrm{R}^{\text{lower}}$ ***fails M6 (observational).*** *Merging two groups of sacrifice events changes per-capability attributions under max-attribution: the joint maximum over all events is less than the sum of within-group maxima. This failure is about the* lower bound*, a weaker claim—a lower bound failing superadditivity does not imply the underlying quantity fails superadditivity. Example 2 witnesses this.*

Example 1 ($\mathrm{vol_R}$ is invariant under removal of unexercised capabilities)**.** *Consider a single-agent system with two poset-independent capabilities:*

- *c_A = "drive to gas station α", $w(c_A) = 1$.*
- *c_B = "drive to gas station β" (different location), $w(c_B) = 1$.*

The system poset is $P = \{c_A, c_B\}$ with $\mathrm{vol_P}(P) = 2$. c_A and c_B are poset-independent: different fuel-access pathways with no shared participants.

Under the exercise indicator of Appendix A, the agent uses only c_A (gas station α is closer), so $P_W^{\text{ex}} = \{c_A\}$ and $\mathrm{vol_R}(P) = \mathrm{vol_P}(P_W^{\text{ex}}) = 1$. (Note: c_B is not *load-bearing at any time in W—removing it leaves all realized outputs unchanged.)*

Now remove c_B: the system becomes $P' = \{c_A\}$ with $\mathrm{vol_P}(P') = 1$ (so $\mathrm{vol_P}$ *contracts by $w(c_B) = 1$). The agent's behavior is unchanged, so $P_W^{\text{ex}} = \{c_A\}$ and $\mathrm{vol_R}(P') = 1$, unchanged.*

The measure $\mathrm{vol_R}$ *is therefore* invariant *under the removal of the unexercised c_B, while* $\mathrm{vol_P}$ *contracts by $w(c_B)$. A* $\mathrm{vol_R}$*-maximizing actor would face no penalty for shedding unexercised capabilities; a* $\mathrm{vol_P}$*-maximizing actor would pay $w(c_B)$.*

This is not a counterexample to M6 as a measure axiom: for the fixed exercised sub-poset $P_W^{\text{ex}} = \{c_A\}$, $\mathrm{vol_R}$ *trivially satisfies M6 (and all other* $\mathrm{vol_P}$*-inherited axioms by restriction). The failure is at the level of the* mapping *$\mathcal{P} \mapsto \mathrm{vol_R}^{\mathcal{P}}$: this mapping is not monotone under capability-set expansion ($P \supset P'$ but $\mathrm{vol_R}(P) = \mathrm{vol_R}(P')$). Equivalently, the mapping is not strictly contracted by capability removal when the removed capability is unexercised. This non-invariance is what breaks the self-balancing transfer (Corollary 2).*

Example 2 ($\mathrm{vol}_\mathrm{R}^{\text{lower}}$ M6-failure via max-attribution)**.** *Consider two sacrifice events with the same bundle $Y = \{a, b\}$ but different decomposition coefficients (permitted under S5, since event-local decomposition is fixed at observation time):*

- *Event E_1: $\Delta\mathrm{vol_P}(X_1) = 1$, $\alpha_{1,a} = 0.5$, $\alpha_{1,b} = 0.5$.*
- *Event E_2: $\Delta\mathrm{vol_P}(X_2) = 1$, $\alpha_{2,a} = 0.3$, $\alpha_{2,b} = 0.7$.*

Let group $A = \{E_1\}$ and group $B = \{E_2\}$. Under Theorem 2 Part B:

$$\mathrm{vol}_\mathrm{R}^{\text{lower}}(a \mid A) = 0.5, \qquad \mathrm{vol}_\mathrm{R}^{\text{lower}}(b \mid A) = 0.5,$$
$$\mathrm{vol}_\mathrm{R}^{\text{lower}}(a \mid B) = 0.3, \qquad \mathrm{vol}_\mathrm{R}^{\text{lower}}(b \mid B) = 0.7.$$

Summing per-capability: $\mathrm{vol}_\mathrm{R}^{\text{lower}}(A) = 1.0$, $\mathrm{vol}_\mathrm{R}^{\text{lower}}(B) = 1.0$.

In the merged group $A \cup B$, max-attribution over both events: $\mathrm{vol}_\mathrm{R}^{\text{lower}}(a \mid A \cup B) = \max(0.5, 0.3) = 0.5$ *and* $\mathrm{vol}_\mathrm{R}^{\text{lower}}(b \mid A \cup B) = \max(0.5, 0.7) = 0.7$. *Total:* $\mathrm{vol}_\mathrm{R}^{\text{lower}}(A \cup B) = 1.2$.

M6 (superadditivity) predicts $\mathrm{vol}_\mathrm{R}^{\text{lower}}(A \cup B) \geq \mathrm{vol}_\mathrm{R}^{\text{lower}}(A) + \mathrm{vol}_\mathrm{R}^{\text{lower}}(B) = 2.0$*, but* $\mathrm{vol}_\mathrm{R}^{\text{lower}}(A \cup B) = 1.2 < 2.0$*. M6 fails for* $\mathrm{vol}_\mathrm{R}^{\text{lower}}$ *in the subadditive direction—max-attribution over merged* *events takes the larger single contribution per capability rather than summing across groups, so the joint aggregate is bounded above by the sum of within-group aggregates.*

This is a property of the lower-bound construction*, not of the underlying* $\mathrm{vol}_{\mathrm{R}}$*: the true* $\mathrm{vol}_{\mathrm{R}}$ *aggregate of the merged events need not be subadditive, but our observational lower bound is. The non-self-balancing corollary below therefore relies on the capability-removal non-invariance of M6(a), not on this lower-bound subadditivity.*

Corollary 2 ($\mathrm{vol}_{\mathrm{R}}$ Is Not Self-Balancing)**.** *The self-balancing property does* not *transfer from* $\mathrm{vol}_{\mathrm{P}}$ *to* $\mathrm{vol}_{\mathrm{R}}$*. Paper 2's self-balancing theorem [Lasser, 2026f, Theorem 1] relies on the fact that* any *capability removal contracts* $\mathrm{vol}_{\mathrm{P}}$*, making* $\mathrm{vol}_{\mathrm{P}}$*-maximization incompatible with unilateral capability reduction. For* $\mathrm{vol}_{\mathrm{R}}$*, removing an unexercised capability leaves the measure unchanged (Example 1), so a* $\mathrm{vol}_{\mathrm{R}}$*-maximizer can shed unused capabilities without penalty. The protective coupling between measure-maximization and capability preservation that* $\mathrm{vol}_{\mathrm{P}}$*'s self-balancing encodes therefore does not apply to* $\mathrm{vol}_{\mathrm{R}}$*.*

The corollary does not *follow from* $\mathrm{vol}_{\mathrm{R}}$ *failing M6 as a measure axiom—it does not (Theorem 3, M6(a))—nor from* $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$ *failing M6 alone (a lower bound failing measure axioms is unremarkable). The corollary rests on the system-level non-invariance of M6(a):* $\mathrm{vol}_{\mathrm{R}}$*'s value depends on which capabilities are exercised, and unexercised capabilities can be removed without affecting that value.*

This is the structural reason to use $\mathrm{vol}_{\mathrm{R}}$ as a diagnostic rather than an objective. An actor maximizing $\mathrm{vol}_{\mathrm{R}}$ would lack the automatic diversity-preservation that makes $\mathrm{vol}_{\mathrm{P}}$ safe: exercise is rewarded, but dormant optionality is not protected.

Remark 12 (When M6 does hold for $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$)**.** *M6 holds for* $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$ *when the merged groups have sacrifice events in poset-independent bundle categories (Definition 10)—each unbenchmarked capability appears in sacrifice events from only one group, and cross-group capabilities are poset-independent. Under this non-redundancy condition,* $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$ *inherits M1–M6. Even so, the self-balancing property transfers to* $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$ *only as a* lower bound *on self-balancing—*$\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$ *being self-balancing does not imply* $\mathrm{vol}_{\mathrm{R}}$ *itself is self-balancing.*

6 Commitment-Layer Composition

6.1 Motivation

The revealed-sacrifice channel observes trade events. In its raw form, each event exposes (i, X, Y, t)—the agent's identity, what they surrendered, what they acquired, and when. This is already more private than direct $\mathrm{vol}_{\mathrm{R}}$ measurement, but the framework can do better: the commitment protocol of [Lasser, 2026a, Definition 6] allows the sacrifice event to be committed with hiding, so that only the $\mathrm{vol}_{\mathrm{R}}$-relevant content (the magnitude $\Delta\mathrm{vol}_{\mathrm{P}}(X)$ and the category of Y) is revealed.

6.2 Formal construction

Definition 9 (Committed Revealed-Sacrifice Event)**.** *A committed revealed-sacrifice event is a tuple* (Γ, π, t) *where:*

- $\Gamma = \mathrm{Commit}((\Delta\mathrm{vol}_{\mathrm{P}}(X), \mathrm{category}(Y)), r)$ *is a commitment to the sacrifice magnitude and bundle category, using randomness r [Lasser, 2026a, Definition 6];*
- π *is a zero-knowledge proof [Lasser, 2026a, Definition 7] that:*
  - *(a) the commitment* Γ *corresponds to a valid revealed-sacrifice event satisfying the objectively verifiable components: S3 (bundle coherence) and the objective component of S2;*
  - *(b)* $\Delta\mathrm{vol}_{\mathrm{P}}(X) \geq 0$*;*
  - *(c) the trade actually occurred (linked to a verifiable ledger entry);*
  - *(d) the bundle Y lies wholly within the claimed partition component:* $Y \subseteq C_{\mathrm{category}(Y)}$ *(Definition 10(ii));*

- *t is the event timestamp (public).*

The committed event reveals $\Delta\mathrm{vol}_\mathrm{P}(X)$ (the sacrifice magnitude) and $\mathrm{category}(Y)$ (the bundle's partition label). It hides agent identity i, the specific benchmarked capability X, the specific unbenchmarked bundle Y, and the counterparty.

Remark 13 (S0, S1, S4(a), and S5 are not ZK-verifiable)**.** *Assumptions S0 (calibration), S1 (free choice), S4(a) (valuation additivity), and S5 (decomposition validity) are about the agent's internal state or the true vol_R distribution. These cannot be verified in zero knowledge [Goldwasser et al., 1985] without an oracle for the agent's subjective state. The ZK proof verifies the* objective *conditions (S2, S3, and S4(b) via poset structure); the* subjective *conditions are structural assumptions about the trade environment, not properties the commitment protocol can enforce.*

Definition 10 (Category Partition)**.** *The function* $\mathrm{category}(Y)$ *maps each unbenchmarked bundle to a label in a* public partition $\mathcal{C} = \{C_1, \dots, C_m\}$ *of the unbenchmarked capability space U, subject to two conditions:*

(i) Poset-independence: *for any $j \neq k$, no capability in C_j is related by subsumption or cooperative composition to any capability in C_k.*

(ii) Bundle containment: *each bundle Y lies wholly within a single partition component:* $\mathrm{category}(Y) = j$ *iff $Y \subseteq C_j$.*

The partition is published as part of the framework's measurement infrastructure and is fixed for a given trade window. This definition is load-bearing for aggregation: distinct category labels imply poset-independence of the underlying bundles (Remark 8), which is what Theorem 2 Part A requires.

6.3 Computability of the Part A formula from committed events

Proposition 6 (Part A Formula is Computable from Committed Events on Distinct-Category Batches)**.** *For a batch of committed revealed-sacrifice events (Definition 9) whose revealed category labels $\{\mathrm{category}(Y)[_n]\}$ are* pairwise distinct*, the* formula *of Theorem 2 Part A—the right-hand sum $\sum_n \Delta\mathrm{vol}_\mathrm{P}(X_n)$ together with the poset-independence check between bundles—is evaluable without opening the hidden fields: the magnitudes $\Delta\mathrm{vol}_\mathrm{P}(X_n)$ and labels $\mathrm{category}(Y)[_n]$ are revealed, and distinct labels imply poset-independent bundles via Definition 10(i). Batches with colliding labels fall outside this certificate and must be handled separately (Remark 15).*

This is a computability *result, not an* admissibility *result. Applying Theorem 2 Part A to a batch of committed events additionally requires that the theorem's hypotheses hold—the genuine-exchange conditions (i)–(iv) of Definition 1 and assumptions S0, S1, S2, and S4(a). The ZK proof π of Definition 9 certifies only the objectively verifiable subset (S2's magnitude component, S3 for Part B uses, and the bundle-containment check $Y_n \subseteq C_{\mathrm{category}(Y)[_n]}$). The remaining hypotheses—S0 (valuation-bounded-by-exercise), S1 (free choice), S4(a) (valuation additivity), and the cooperative-closure conditions (iii)–(iv)—are about agent-internal state or poset-structural facts the commitment protocol cannot itself attest (Remark 13). Their validity must be established externally: by institutional filters on the trade environment, by sampled audit, or by assumption at the batch level.*

Proof. We establish formula computability; admissibility is discussed in the statement and is not claimed here.

(i) Sacrifice magnitudes. By Definition 9, each committed event reveals $\Delta\mathrm{vol}_\mathrm{P}(X_n)$ directly, so the aggregate $\sum_n \Delta\mathrm{vol}_\mathrm{P}(X_n)$ is evaluable without opening the commitment. (In the hiding variant of Proposition 7, only B and a ZK attestation $\sum_n \Delta\mathrm{vol}_\mathrm{P}(X_n) \geq B$ are published; both forms expose the aggregate at the intended privacy level.)

(ii) Pairwise poset-independence of bundles (distinct labels). By Definition 10(i), any two distinct partition components C_j and C_k are poset-independent: no subsumption or cooperative-composition link connects any $c \in C_j$ to any $c' \in C_k$. Definition 9(d)'s ZK proof attests that $Y_n \subseteq C_{\mathrm{category}(Y)[_n]}$, so Y_n lies wholly within a single component. Under the pairwise-distinct-label hypothesis, the bundles lie in distinct components and inherit poset-independence from the partition. Both checks use only revealed or ZK-attested fields.

When two events share the same category label $\mathrm{category}(Y)[_n] = \mathrm{category}(Y)[_{n'}]$, both bundles lie inside the same partition component C_j; poset-independence between $Y_n, Y_{n'} \subseteq C_j$ is a finer fact about sub-structure within C_j that Definition 9 does not attest. The distinct-label certificate is therefore a *sufficient* certificate for Part A's poset-independence hypothesis, not a *necessary* one; see Remark 15 for how to handle batches with colliding labels.

(iii) What is not certified. Neither the above computation nor the ZK proof π establishes the genuine-exchange conditions (iii)–(iv) (cooperative-closure across $X/H \setminus X$ and $Y/H \setminus X$), S0 (agent's valuation bounded by realized exercise), S1 (voluntariness and non-degradation), or S4(a) (valuation additivity over bundles). These depend on facts about the agent's internal state, the pre-trade holding set, and the poset structure that are not exposed by the commitment. The Part A bound is admissible only when those are established by an external mechanism (Remark 14). ■

Remark 14 (External attestation model)**.** *In practice, the non-ZK-verifiable hypotheses are handled at the institutional layer: (a) the trade environment is filtered so only trades satisfying S1's non-degradation are admitted as sacrifice events; the need-sufficiency architecture developed in the companion paper [Lasser, 2026d] supplies the structural filter and polarity-boundary mechanism for this; (b) the category partition $\mathcal{C}$ is curated so cooperative-closure conditions (iii)–(iv) hold structurally; (c) S0 and S4(a) are assumed at the batch level with periodic sampled-audit reconciliation. The commitment layer is orthogonal: it enforces privacy and certifies the objective facts, but does not and cannot certify the behavioural and structural assumptions.*

Remark 15 (Handling category-label collisions)**.** *When two committed events share a category label, Proposition 6's distinct-label certificate does not apply to that pair: both bundles lie in the same partition component C_j, and poset-independence between same-component subsets is finer structure than the commitment interface exposes. Three operational responses are available without weakening the theorem:*

- *(a)* Sub-batch partitioning. *Partition the batch into maximal sub-batches whose labels are pairwise distinct and compute Part A separately per sub-batch. The resulting lower bounds are valid for each sub-batch's covered subset; the full-batch aggregate is the sum across sub-batches only when the sub-batches are themselves poset-independent, which is guaranteed when sub-batches are built from distinct partition components. A single-component batch with multiple events yields at most the per-event Part A bound,* $\mathrm{vol}_{\mathrm{R}}(Y_n) \geq \Delta\mathrm{vol}_{\mathrm{P}}(X_n)$ *for each n; aggregation requires the additional independence.*
- *(b)* Finer partition. *Refine the public partition $\mathcal{C}$ to sub-components such that typical batch collisions become distinct-label pairs. This is a governance choice trading category-granularity against commitment complexity.*
- *(c)* Richer commitments. *Extend Definition 9 to attest explicit subset-disjointness or cooperative-closure of $Y_n, Y_{n'}$ within their shared component C_j. This is the path Part B's per-capability certificate (Remark 16) also walks.*

The paper's stated results use option (a) implicitly: Proposition 7's ZK-aggregation rollup explicitly proves pairwise-distinct labels as part of its (ii), matching the distinct-batch certificate of Proposition 6.

Remark 16 (Part B requires additional committed fields)**.** *Part B's per-capability refinement requires $\alpha_{n,c}$ coefficients and stable capability identities across events. The committed interface reveals only category labels. For Part B to operate on committed data, the commitment must additionally expose a per-capability coordinate system within each category component and the corresponding α values. This is a richer commitment that reveals more structure than Part A requires.*

Proposition 7 (ZK Aggregation over Hiding-Committed Events)**.** *The Pedersen commitment scheme underlying Definition 9 is hiding [Lasser, 2026a, Definition 6] by default; the accompanying disclosure of* $(\Delta\mathrm{vol}_{\mathrm{P}}(X_n), \mathrm{category}(Y)[_n])$ *is an* optional *publication choice. Consider the* hiding variant *of the committed-event interface in which each event publishes* (Γ_n, π_n, t_n) *but does not release the pair* $(\Delta\mathrm{vol}_{\mathrm{P}}(X_n), \mathrm{category}(Y)[_n])$ *in cleartext; the ZK proof π_n is strengthened to additionally attest that* $\mathrm{category}(Y)[_n] \in \{1, \ldots, m\}$ *(a valid partition label).*

Fix a public upper bound $N_{\max}$ on batch size. Given a batch of at most $N_{\max}$ hiding-variant committed events, a rollup proof can establish:

(i) $\sum_n \Delta\mathrm{vol}_{\mathrm{P}}(X_n) \geq B$ *(aggregate bound);*

(ii) the bundle category labels $\{\mathrm{category}(Y)[_n]\}$ *are pairwise distinct (as a multiset over the* $N_{\max}$ *positions, with unused positions filled by a distinguished "null" label);*

for a public bound B*, without revealing the individual* $\Delta\mathrm{vol}_{\mathrm{P}}(X_n)$*, the specific category labels, or the true batch size* $N \leq N_{\max}$*.*

By Definition 10, distinct category labels correspond to poset-independent partition components, so (ii) establishes the poset-independence required by Theorem 2 Part A. Both (i) and (ii) are formula inputs *that the ZK rollup certifies; the theorem's admissibility hypotheses—the genuine-exchange conditions of Definition 1 and assumptions S0, S1, S2, S4(a)—remain outside the ZK layer (Remark 13) and must be supplied through external attestation (Remark 14).*

Proof. The prover holds the full set of N committed events together with their openings $(\Delta\mathrm{vol}_{\mathrm{P}}(X_n), \mathrm{category}(Y)[_n], r_n)$. To standardize batch size the prover pads with $N_{\max} - N$ "null" events committing to $(0, \mathrm{null})$ with fresh randomness. Padding is homomorphically invisible—the contribution to the aggregate sum is zero and the null label is distinguished from all partition components—so the extended batch has the same aggregate and the same non-null category multiset.

Component (i)—range proof on the aggregate sum. Using a standard Pedersen commitment range proof [Lasser, 2026a, Definition 7], the prover demonstrates $\sum_{n=1}^{N_{\max}} \Delta\mathrm{vol}_{\mathrm{P}}(X_n) \geq B$. The summation commitment is the product $\prod_n \Gamma_n$, which binds to the aggregate by the additive-homomorphic property of Commit. Null events contribute 0 so the sum equals $\sum_{n=1}^{N} \Delta\mathrm{vol}_{\mathrm{P}}(X_n)$. The proof reveals neither the individual $\Delta\mathrm{vol}_{\mathrm{P}}(X_n)$ nor the true N (the verifier sees only a length-$N_{\max}$ vector of commitments).

Component (ii)—pairwise-distinct non-null labels. The prover commits to the extended label vector $(\mathrm{category}(Y)[_1], \ldots, \mathrm{category}(Y)[_{N_{\max}}])$, where positions corresponding to null events hold the null label. The ZK argument shows (a) each non-null $\mathrm{category}(Y)[_n]$ opens to a valid leaf of the public partition Merkle root $\mathcal{C}$, and (b) the multiset of non-null labels contains no duplicates. Duplicate-freeness is a standard ZK set-disjointness argument: commit to a sorted copy, prove strict-monotone ordering among non-null elements, and prove the permutation relation to the unsorted commitments via a shuffle argument [Lasser, 2026a, Definition 7].

Formula-input certification. By Definition 10(i), pairwise-distinct non-null labels imply pairwise poset-independent bundles. Combined with (i), the rollup certifies the two ZK-verifiable inputs to the Part A formula (Proposition 6): the aggregate sacrifice magnitude meets the threshold B, and the bundles are pairwise poset-independent. The verifier learns only the public parameters $(B, N_{\max}, \mathcal{C})$ and the length-$N_{\max}$ commitment vector.

Admissibility is not ZK-certified. The inference ${\mathrm{vol}_{\mathrm{R}}}^U \geq B$ requires, in addition, that Theorem 2 Part A's non-ZK-verifiable hypotheses hold across the batch: the genuine-exchange conditions (i)–(iv) of Definition 1, S0, S1, S2, and S4(a). These are not part of the rollup proof (Remark 13). Under external attestation of those hypotheses (Remark 14), the hidden-batch rollup certifies ${\mathrm{vol}_{\mathrm{R}}}^U \geq B$; without such attestation, the rollup certifies only that the Part A formula's inputs meet the stated thresholds while hiding individual $\Delta\mathrm{vol}_{\mathrm{P}}(X_n)$, specific category labels, and the true N. ■

Remark 17 (Hiding variant is strictly richer than Definition 9)**.** *The hiding-variant interface is a strict privacy upgrade: it hides what Definition 9 discloses. The default interface suffices for routine operational accounting where committed events are published to a public ledger; the hiding variant is useful when a higher-level aggregator publishes only batch-level attestations to downstream verifiers. Both interfaces rely on the same Pedersen commitment primitive; they differ only in which openings are released alongside the commitment.*

7 Integration: Duality and the Shared Privacy Discipline

7.1 The complementary pair

	Controlled relaxation [Lasser, 2026h]	**Revealed sacrifice** (this paper)
Question	Is the restriction warranted?	Is the capability valuable?
Evidence	Framework-initiated exercise	Agent-initiated sacrifice
Direction	Framework injects observation	Framework receives disclosure
Scope	Bounded sub-poset S, bounded τ	Bounded to the trade event
Privacy cost	Scope-limited monitoring	None beyond what the trade creates
Adversarial model	Agent must tolerate test	Agent must operate (buy, work)
Dimension	Risk (restriction lattice)	Value (experiential $\mathrm{vol}_{\mathrm{R}}$)

Together they cover both axes of the preemptive-restriction criterion of [Lasser, 2026c, Proposition 1]: restriction cost (controlled relaxation tests the risk side) and experiential value (revealed sacrifice bounds the value side). Controlled relaxation tells us whether a feared harm is real; revealed sacrifice tells us whether the forgone capability is worth recovering. Without both, the restriction decision is underdetermined.

7.2 The shared privacy discipline

Three papers in the sequence [Lasser, 2026e,h] and the present paper share a structural discipline: **resolution through frequency, not through depth.**

- The phase boundary of [Lasser, 2026e, Theorem 2]: $\rho^{\mathrm{cross}}_{\mathrm{min}}$ counts observation *channels*, not depth of observation per channel.
- Controlled relaxation [Lasser, 2026h]: more precision on risk claims comes from running more tests (Lasser, 2026h, Theorem 2, convergence depends on test count), not from surveilling each test more intrusively.
- Revealed sacrifice (this paper): more precision on $\mathrm{vol}_{\mathrm{R}}$ comes from observing more trades (Propositions 3 and 2), not from observing each trader more intrusively. The commitment layer (Section 6) ensures that even the trade-level observation is privacy-minimal.

This shared discipline is the sequence's answer to the panopticon temptation: the framework gains measurement capability through *more independent signals*, not through *deeper access to any single signal*. This is consonant with the cross-substrate redundancy principle of [Lasser, 2026e, Theorem 2] and the cryptographic opacity commitment of [Lasser, 2026a, Definition 6].

7.3 Connection to the phase-boundary analysis

The phase-boundary analysis of [Lasser, 2026e] assumes $\mathrm{vol}_{\mathrm{P}}$ accurately measures the quantity the framework optimizes. The present paper asks whether that assumption holds. When β^{lower} is low, the phase-boundary parameters (r_{S}, r_{W} from [Lasser, 2026e, Theorem 1]) may be miscalibrated because the actor is optimizing a proxy ($\mathrm{vol}_{\mathrm{P}}$) that diverges from the true target.

A natural extension (not pursued here): extend the phase-boundary analysis to incorporate the $\mathrm{vol}_{\mathrm{P}}/\mathrm{vol}_{\mathrm{R}}$ divergence. The effective self-correction rate $r^{\mathrm{eff}}_{\mathrm{S}}$ would depend on β^{lower}.

7.4 Bridge to the companion paper

This paper establishes the lower-bound measurement theorem; the companion paper [Lasser, 2026d] completes the architecture on three fronts that this paper takes as external preconditions or defers as diagnostics:

- **S1 admissibility filter.** Theorem 2 requires above-sufficiency sacrifice events; below-sufficiency sacrifices have reversed polarity and fail S1. The *need-sufficiency architecture* [Lasser, 2026d, Section 2] establishes a structural poset-construction precondition that identifies the polarity boundary and filters below-sufficiency events before they reach the observation channel.

- **Gap decomposition and diagnostic interpretation.** A low β^{lower} has multiple possible causes (restricted capabilities, uncovered want-space, structural residual). [Lasser, 2026d, Section 3] partitions the gap into four disjoint cells—restricted, covered, dormant, residual—and supplies an $O(|P|+E+M)$ computability proposition under poset-boundary closure.
- **Wireheading-consistent signature and residual class.** [Lasser, 2026d, Sections 4–5] develops the Herfindahl–Hirschman-Index-based concentration signal from trade-flow data and characterizes the residual class (capabilities structurally outside the sacrifice channel's reach) via a revealed-observation carve that shifts the residual class from a measurement-apparatus question to a channel-reach question.

Together, the two papers are the measurement theorem (this paper) and the diagnostic architecture that interprets what the measurement shows [Lasser, 2026d].

8 Discussion and Open Questions

8.1 What this paper establishes

Direct vol_R measurement is privacy-incompatible. The framework's commitments in [Lasser, 2026a,e,h] rule out the panopticon-style observation that direct measurement would require. This is a structural finding, not a practical limitation.

Revealed sacrifice is a privacy-minimal surrogate. The sacrifice channel satisfies three formal privacy properties (Theorem 1) plus two institutional preconditions (Proposition 1), and composes with the commitment layer of [Lasser, 2026a] (Propositions 6 and 7) to produce a privacy-minimal observation mechanism on distinct-category batches.

The B-to-C gap is characterized by trade data. The aggregate lower bound (Theorem 2) lower-bounds the exercised fraction β (equivalently, upper-bounds the B-to-C gap $1-\beta$) on the observed subset, under S0–S2 and S4(a). The constructed lower bound is non-decreasing in observed admissible events: Part A via sub-batch supremum (Proposition 3) and Part B via per-capability max-attribution (Proposition 2).

Two sacrifice channels, substitutable with residuals. The money and time channels are substitutable via cooperative markets (employment and hiring) and triangulate the same underlying sacrifice substrate. The joint aggregate weakly dominates either channel alone unconditionally; strict domination obtains at the non-fungibility residuals (agent-specific time, capital-concentrated access) when the trade window contains an event in the relevant residual category, per Corollary 1.

vol_R is not self-balancing. vol_R exhibits non-invariance under capability removal: eliminating an unexercised capability leaves vol_R unchanged while vol_P contracts (Theorem 3, M6(a); Example 1). The self-balancing protection vol_P enjoys—every removal contracts the measure—therefore does not transfer to vol_R (Corollary 2). vol_R is a diagnostic, not an objective. This structural finding is independent of the observation mechanism.

Complementarity with controlled relaxation. Together, this paper and [Lasser, 2026h] cover the risk and value dimensions of the preemptive-restriction criterion of [Lasser, 2026c, Proposition 1]. Neither is sufficient alone.

8.2 Limitations

Assumption load. The full per-capability refinement (Theorem 2 Part B) requires S0–S5 plus within-bundle poset-independence (Definition 4). S0 (calibration) and S1 (free choice) are the most empirically demanding, and neither is ZK-verifiable (Remark 13). Part A (S0–S2 and S4(a) only) is the more defensible result; Part B should be understood as the refinement available under stronger conditions.

B-to-C gap inheritance. The damage bound of [Lasser, 2026h, Theorem 1] bounds contraction of $\mathrm{vol}_{\mathrm{P}}$, not of the experiential target directly. The present paper provides a lower bound on $\mathrm{vol}_{\mathrm{R}}$ as a diagnostic, but the diagnostic is itself subject to the B-to-C gap on the benchmarked side: the sacrifice signal $\Delta\mathrm{vol}_{\mathrm{P}}(X)$ is only as accurate as $\mathrm{vol}_{\mathrm{P}}$'s measurement of the benchmarked capability.

External admissibility layer. Several non-ZK-verifiable hypotheses require institutional attestation to hold operationally: S0 (valuation bounded by exercise), S1 (free choice and non-degradation), S4(a) (valuation additivity), and the genuine-exchange conditions (iii)–(iv) of Definition 1. Remark 14 describes the operational shape of this attestation layer. The specific filter for S1's non-degradation condition—rejecting below-sufficiency sacrifices as structurally involuntary—is developed in the companion paper [Lasser, 2026d] and treated here as an external precondition.

8.3 Open questions

1. **Hedonic disaggregation of bundles.** Theorem 2 Part B requires event-local decomposition coefficients $\alpha_{n,c}$. Classical hedonic regression [Rosen, 1974] handles the money channel. The time-sacrifice channel needs an analogous framework: how does one decompose a block of time spent on a complex activity (parenting, which involves teaching, emotional regulation, physical care, relationship-building) into per-capability contributions?
2. **Non-stationarity of revealed preferences.** Preferences shift over time. The lower bound from past trades may overstate $\mathrm{vol}_{\mathrm{R}}$ at present. An EWMA decay paralleling the risk-trust dynamics of [Lasser, 2026g, Definition 4] seems natural but requires empirical calibration.
3. **Coerced sacrifice.** Assumption S1 (free choice) is load-bearing. Under duress, the agent's trade does not reveal preferences. Filtering coerced sacrifice events may require commitment-based attestation of choice sets [Lasser, 2026a].
4. **Market-failure cases.** In contexts without functioning markets (pre-capitalist societies, closed economies, intra-family sharing), the money-sacrifice channel is quiet even though $\mathrm{vol}_{\mathrm{R}}$ activity is real. Does the time-sacrifice channel alone suffice, or does the framework need a non-market sacrifice variant?
5. **Commitment infrastructure at scale.** A fully committed trade ledger at the scale of a modern economy is an infrastructure problem. The privacy guarantee (Section 6) is conditional on the commitment infrastructure existing, which is the remit of [Lasser, 2026a].
6. **Population-level calibration (S0).** Assumption S0 requires that individual agents do not overvalue unbenchmarked bundles. A robust version of Theorem 2 would replace S0 with a weaker population-level condition: $E[U_i(Y)] \leq \mathrm{vol}_{\mathrm{R}}(Y) + \epsilon$ for some bounded bias ϵ. Specific behavioural mechanisms that stress individual-agent S0 and motivate a population-level weakening include: (i) *ritualized payment*, where a transaction's nominal price is symbolic rather than exchange-valued (a token fee to "soften" receiving something for free); (ii) *gift aversion*, where agents refuse free offers because they interpret absence of price as signalling hidden defect, inflating observed sacrifices on nominally free goods; (iii) *time-for-money inefficiency*, where a given hour's wage rate r_i does not reflect the agent's marginal valuation (labour-market frictions, status wages, family-care default allocations). Each of these produces systematic biases on observed sacrifice magnitudes that the per-agent S0 cannot absorb; a population-level reformulation that lets such biases cancel (or bounds the residual) is the natural follow-up.

Further open questions in the companion paper. Open questions on threshold governance, need identification, capability-stuffing as structural avoidance, and the full architecture of gap-decomposition governance are collected in the companion paper [Lasser, 2026d].

Author Contributions

Teague Lasser identified the B-to-C gap as a structural open problem in the GFM sequence and directed the reframing from direct $\mathrm{vol}_{\mathrm{R}}$ measurement to revealed-sacrifice observation. Directed the formal development of the two-channel (money and time) sacrifice model, the aggregate lower bound theorem, and the commitment-layer composition. Identified the duality between controlled relaxation and revealed sacrifice (risk dimension versus value dimension) and the shared privacy discipline (resolution through frequency, not depth). Made final editorial decisions and is responsible for the paper's intellectual direction and all claims made.

Claude Opus 4.7 (Anthropic) drafted the formal exposition: the revealed-sacrifice observation model with its three formal privacy properties and two institutional preconditions (Section 2), the aggregate B-to-C lower bound theorem with its two-level structure (Section 3), the two-channel sacrifice analysis (Section 4), the axiom-inheritance analysis and non-self-balancing corollary (Section 5), and the commitment-layer composition including the ZK-aggregation proposition (Section 6).

GPT 5.4 (OpenAI) served as cold technical reviewer across thirty review cycles, catching formal gaps across all sections—including the requirement for pairwise poset-independence (not merely set-disjointness) in the theorem's aggregation step, the distinction between $\mathrm{vol}_{\mathrm{R}}$ being non-invariant under capability removal and a measure-axiom failure in the non-self-balancing finding, the distinct-label scope of the commitment rollup, the full poset-boundary closure (cooperative + subsumption) required by the combined diagnostic, and the scope-mismatch between Part A and Part B monotonicity claims.

Transparency note. Both AI systems operated as tools under human direction. Neither system has continuity across sessions, cannot take responsibility for the work in the sense required by most venue authorship policies, and cannot respond to reviewer queries independently. They are listed as authors to accurately represent their contributions to the intellectual content of the paper, not to claim that they meet all criteria of traditional academic authorship. The corresponding author for all inquiries is Teague Lasser.

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A The Exercise Indicator (Supplementary)

A.1 Scope limitation

The exercise indicator is retained as a *supplementary* measurement tool, applicable only where interior access is legitimate—specifically, within the collective's own operational perimeter where no privacy issue arises (e.g., the framework measuring the exercise status of its own internal components, or a cooperative measurement within a consenting sub-collective).

Definition 11 (Exercise Indicator, Internal Perimeter)**.** *Let G be a capability collective operating within an agreed-upon observation perimeter $O \subseteq P$. We require O to be* boundary-closed in the poset-independence sense of Definition 10(i)*: neither cooperative-composition nor subsumption links cross the $O/P \setminus O$ partition. Concretely, two conditions:*

- *(BC1)* Cooperative closure. *For every cooperative capability $z \in P$, either all participants of z lie in O or none do (so cooperatives are not split across the boundary).*
- *(BC2)* Subsumption closure. *No subsumption relation crosses the boundary: if $c \succeq c'$ in P (i.e., c subsumes c'), then either both $c, c' \in O$ or both $c, c' \in P \setminus O$.*

Together (BC1) and (BC2) make O and $P \setminus O$ poset-independent in the sense of Definition 10(i), which is what licenses additive composition across the boundary below. (BC1) alone—the weaker cooperative-only condition—is insufficient: subsumption edges crossing the boundary would break M4 on the $O/P \setminus O$ partition.

For each capability $d \in O$, define:

$$e_t(d) = \begin{cases} 1 & \text{if there exists a realized cooperative output in } [t-\Delta, t] \\ & \text{such that removing } d \text{ from } O \text{ makes the output unrealizable,} \\ 0 & \text{otherwise.} \end{cases}$$

The indicator is scoped to the observation perimeter O. The framework may use $e_t(d)$ for internal diagnostics within O, but may not *apply it to capabilities outside O (where privacy constraints apply).*

Remark 18 (The two channels are complementary)**.** *Within the perimeter O, the exercise indicator provides exact exercise status ($e_t(d) \in \{0, 1\}$). Outside O, the sacrifice channel provides a lower bound. Under the boundary-closed hypothesis of Definition 11, the aggregate $\mathrm{vol}_{\mathrm{R}}$ diagnostic is a valid lower bound obtained by additive composition:*

$$\mathrm{vol}_{\mathrm{R}}^{\mathrm{combined}} \;:=\; \mathrm{vol}_{\mathrm{R}}^{\mathrm{exact}}(O) + \mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(P \setminus O) \;\leq\; \mathrm{vol}_{\mathrm{R}}(O) + \mathrm{vol}_{\mathrm{R}}(P \setminus O) \;=\; \mathrm{vol}_{\mathrm{R}}(P),$$

where the final equality uses axiom M4 applied to the poset-independent pair O, $P \setminus O$ (Section 5). $\mathrm{vol}_{\mathrm{R}}^{\mathrm{exact}}(O)$ uses the exercise indicator and $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(P \setminus O)$ uses the sacrifice channel.

Remark 19 (When boundary-closure fails)**.** *If O fails either condition—a cooperative z has participants split across O and $P \setminus O$ (BC1 fails), or a subsumption edge $c \succeq c'$ crosses the boundary (BC2 fails)—then O and $P \setminus O$ are not poset-independent, and M4 additivity across the boundary does not apply. Specifically: under a BC1 failure, z contributes to $\mathrm{vol}_{\mathrm{R}}(P)$ but to neither $\mathrm{vol}_{\mathrm{R}}^{\mathrm{exact}}(O)$ nor $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(P \setminus O)$ individually (the exercise indicator inside O evaluates only capabilities fully within O; the sacrifice channel outside O cannot reach z if z's full participant set is not in $P \setminus O$). Under a BC2 failure, the subsumption edge couples $\mathrm{vol}_{\mathrm{P}}$-mass across the boundary in ways that the independent-component additivity of M4 cannot decompose. In either case $\mathrm{vol}_{\mathrm{R}}^{\mathrm{combined}}$ remains a* conservative *lower bound on $\mathrm{vol}_{\mathrm{R}}(P)$—the subadditivity of $\mathrm{vol}_{\mathrm{P}}$ on non-independent components ensures the sum of pieces is $\leq$ the whole—but the equality with $\mathrm{vol}_{\mathrm{R}}(P)$ above no longer holds. Practical perimeters (the collective's own operational scope, consenting sub-collectives) typically satisfy both boundary-closure conditions because cooperative capabilities and subsumption hierarchies are either wholly internal to the perimeter or wholly external; where either BC1 or BC2 fails, the undercount is quantified by the $\mathrm{vol}_{\mathrm{P}}$-weight of the offending cross-boundary cooperatives and subsumption edges.*

B Proof Sketches and Technical Notes

B.1 Privacy-minimality (Section 2)

Theorem 1 + Proposition 1 strategy. The three formal properties (P1, P3, P4) are proved independently from the channel's construction; the two institutional preconditions (P2, P5) are documented separately in Proposition 1. Proof strategy: P1 (no interior access): by construction of the observation tuple. P3 (discretization): by bounding channel bandwidth with trade frequency. P4 (commitment composability): by composition with [Lasser, 2026a, Definition 6]. P2 and P5 are not theorem content: P2's zero-contribution property is formal (a consequence of the aggregate's summation structure), but the consent interpretation depends on institutional opt-in conventions; P5 depends on public-ledger infrastructure external to the paper's formal scope, and is additionally split between money-channel events (observable from public ledger) and time-channel events (require an additional public wage schedule or imputation rule).

B.2 Aggregate lower bound (Section 3)

Theorem 2 strategy. *Part A (bundle-level, S0–S2 and S4(a), poset-independent bundles, genuine-exchange conditions):*

1. Revealed-preference observation: $U_i[_n]((H_n \setminus X_n) \cup Y_n) \geq U_i[_n](H_n)$.
2. Cancellation to bundle comparison (S4(a) + Definition 1 conditions (i)–(iv), via Lemma 1): $U_i[_n](Y_n) \geq U_i[_n](X_n)$.
3. Calibration chain (S0 + S2), S0 stated directly in window-integrated form: $\mathrm{vol_R}^{[W]}(Y_n) \geq U_i[_n](Y_n) \geq U_i[_n](X_n) \geq \Delta\mathrm{vol_P}(X_n)$. S0's consequent is against $\mathrm{vol_R}^{[W]}$ rather than against the instantaneous $\mathrm{vol}_{\mathrm{R}t_n}$: a sacrifice event records acquisition, not an instantaneous exercise witness, so the bound is about cumulative exercise of Y_n anywhere in W (Definition 3, Section 2.2).
4. Poset-independence of bundles (Definition 10) gives M4 additivity (inheritance proved in Theorem 3 via sub-poset restriction): $\mathrm{vol_R}^{[W]}(\bigcup Y_n) = \sum_n \mathrm{vol_R}^{[W]}(Y_n) \geq \sum_n \Delta\mathrm{vol_P}(X_n)$.

Part B (per-capability refinement, additionally S3–S5 and within-bundle poset-independence; conditional on S5, see Remark 2):

4. Part B is stated for events with $\Delta\mathrm{vol_P}(X_n) > 0$ and bundles Y_n satisfying within-bundle poset-independence (Definition 4, Remark 6); bundles with internal cooperative activation fall back to Part A only. Positivity $\mathrm{vol_R}(Y_n) > 0$ then follows from Part A.
5. Within-bundle poset-independence plus M4 gives $\mathrm{vol_R}(Y_n) = \sum_{c \in Y_n} \mathrm{vol_R}(c)$, making the partition-of-unity constraint $\sum_{c \in Y_n} \alpha_{n,c} = 1$ compatible with S5's share inequality $\alpha_{n,c} \leq \mathrm{vol_R}(c)/\mathrm{vol_R}(Y_n)$ (equality achievable). Combined: $\mathrm{vol_R}(c) \geq \alpha_{n,c} \cdot \Delta\mathrm{vol_P}(X_n)$. S3 (bundle coherence) ensures the hedonic decomposition is non-degenerate so $\alpha_{n,c}$ is well-specified.
6. Per-capability max-attribution across overlapping events (Proposition 2).
7. Summation over capabilities in U (pairwise poset-independent) via a second M4 application.

Key technical challenges:

- *Poset-independence (Part A):* the category partition $\mathcal{C}$ (Definition 10) must be a genuine poset partition. Set-disjointness is necessary but not sufficient (Remark 8).
- *Decomposition validity (S5, Part B):* hedonic regression [Rosen, 1974] produces attribution weights, not $\mathrm{vol_R}$ share lower bounds. S5 is the explicit bridge.
- *Calibration (S0):* the weakest assumption in the chain but the hardest to verify empirically (Open Question 6).

B.3 Axiom inheritance (Section 5)

Theorem 3 strategy. Axiom-by-axiom analysis: M1–M5 are direct from properties of summation/max over non-negative terms (detailed in the body; M4 is proved constructively via sub-poset restriction). M6 for $\mathrm{vol}_{\mathrm{R}}$ as a measure on a fixed exercised sub-poset is inherited from $\mathrm{vol}_{\mathrm{P}}$ by restriction—no axiom failure there. The paper's non-self-balancing claim rests instead on *capability-removal non-invariance*: removing an unexercised capability leaves $\mathrm{vol}_{\mathrm{R}}$ unchanged while $\mathrm{vol}_{\mathrm{P}}$ drops by the removed weight (Example 1). This is a system-level property (of the mapping $\mathcal{P} \mapsto \mathrm{vol}_{\mathrm{R}}{}^{\mathcal{P}}$), not a measure-axiom failure. M6(b) for $\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}$ is witnessed separately by Example 2: max-attribution subadditivity. Corollary 2 follows from (a)'s non-invariance, not from (b)—a lower bound failing measure axioms is unremarkable; what breaks self-balancing transfer is that $\mathrm{vol}_{\mathrm{R}}$ doesn't penalize shedding of unused capabilities.

C Notation Summary

C.1 Symbols introduced in this paper

Symbol	Definition	Ref.
(i, X, Y, t)	Revealed-sacrifice event	Def. 1
$\Delta\mathrm{vol}_{\mathrm{P}}(X)$	$\mathrm{vol}_{\mathrm{P}}$ contribution of surrendered capability	Def. 1
$\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower}}(Y)$	Lower bound on $\mathrm{vol}_{\mathrm{R}}$ of acquired bundle	Def. 1
$[t_0, t_0 + T]$	Aggregate trade window	Def. 2
P_t^{ex}, P_W^{ex}	Exercised sub-poset (instantaneous / window-integrated)	Def. 3
$\mathrm{vol}_{\mathrm{R}t}$, $\mathrm{vol}_{\mathrm{R}}{}^{[W]}$	Realized capability volume (instantaneous / window-integrated)	Def. 3
$\alpha_{n,c}$	Event-local decomposition coefficients	Def. 4
$\mathrm{vol}_{\mathrm{R}}^{\mathrm{lower\,A}}(E)$	Part A lower bound via sub-batch supremum	Prop. 3
$\beta^{\mathrm{lower}}(G, T)$	B-to-C ratio under revealed sacrifice	Def. 6
p	Price (money-sacrifice signal)	Def. 7
r_i	Agent i's market wage rate	Def. 8
(Γ, π, t)	Committed revealed-sacrifice event	Def. 9
$\mathcal{C}$	Public poset-independent partition of U	Def. 10
$\mathrm{category}(Y)$	Bundle's partition label	Def. 10
$e_t(d)$	Exercise indicator (internal perimeter)	Def. 11
O	Observation perimeter	Def. 11
$\mathrm{vol}_{\mathrm{R}}^{\mathrm{exact}}(O)$	Exercise-based realized volume within O	Def. 11
$\mathrm{vol}_{\mathrm{R}}^{\mathrm{combined}}$	Combined diagnostic (exact + lower bound)	App. A

C.2 Assumptions

Assumption	Name	ZK-verifiable?
S0	Valuation-bounded-by-Exercise ($U_i(Y) \leq \mathrm{vol}_{\mathrm{R}}{}^{[W]}(Y)$)	No
S1	Free choice (non-degrading alternative)	Partially
S2	Benchmark grounding ($U_i(X) \geq \Delta\mathrm{vol}_{\mathrm{P}}(X)$)	Objective part only
S3	Bundle coherence (non-degenerate decomposition)	Yes
S4(a)	Valuation additivity	No
S4(b)	$\mathrm{vol}_{\mathrm{R}}$ additivity on poset-independent capabilities	Yes (via poset)
S5	Decomposition validity (α are share lower bounds)	No

C.3 Dependencies on prior papers

Paper	Result	Number	Used in
[Lasser, 2026b]	Population empowerment	Def. 6	§1
[Lasser, 2026b]	Self-balancing	Prop. 1	§1, Cor. 2
[Lasser, 2026f]	Benchmark	Def. 2	Def. 1
[Lasser, 2026f]	Poset measure $\mathrm{vol}_{\mathrm{P}}$	Def. 7	Def. 1, Def. 6
[Lasser, 2026f]	Axioms M1–M6	Prop. 1	Thm. 3
[Lasser, 2026f]	Self-balancing on posets	Thm. 1	Cor. 2
[Lasser, 2026c]	Anti-monopolar property	Prop. 6	§1
[Lasser, 2026c]	Preemptive-restriction criterion	Prop. 1	§7
[Lasser, 2026g]	Observation-window framing	(general)	Def. 2
[Lasser, 2026a]	Commitment Protocol	Def. 6	Def. 9
[Lasser, 2026a]	ZK Capability Proof	Def. 7	Def. 9
[Lasser, 2026e]	Phase boundary	Thm. 1	§7
[Lasser, 2026e]	Channel redundancy	Thm. 2	§7
[Lasser, 2026h]	Controlled relaxation	Def. 3	§7
[Lasser, 2026h]	Damage bound	Thm. 1	§7
[Lasser, 2026d]	Need-sufficiency architecture (S1 filter)	§2	Sec. 2 (as external precondition)
[Lasser, 2026d]	Gap decomposition	§3	§7
[Lasser, 2026d]	Wireheading-consistent signature	§4	§7
[Lasser, 2026d]	Residual class characterization	§5	§7